HCI BOOK SUMMARY Alisa Avdic

CHAPTER 1

Historical context

- Human-computer interaction (HCI) is a field that emerged in the 1980s.
- Prior to the 1980s, computers were too precious and complicated for the average person to use and were mainly used by engineers and scientists.
- Interaction was not a priority for the engineers and scientists who designed and programmed early computers.
- HCI research is important because it underlies advances that migrate into products and processes that people use for work or pleasure.
- This book focuses on research in HCI, specifically experimental research with human participants.
- HCI research is empirical because it is based on observation and experience and is carried out and reported in a manner that allows results to be verified or refuted by other researchers

1.1 Introduction

- HCI emerged in the 1980s and owes a lot to the older discipline of human factors or ergonomics.
- The Association for Computing Machinery Conference on Human Factors in Computing Systems (ACM SIGCHI) is the preeminent annual conference in HCI.
- Human factors is both a science and a field of engineering concerned with human capabilities, limitations, and performance and with designing systems that are efficient, safe, comfortable, and enjoyable for humans.
- HCI is human factors narrowly focused on human interaction with computing technology.
- HCI draws upon interests and expertise in various disciplines such as psychology, sociology, anthropology, cognitive science, computer science, and linguistics.

1.2 Vannevar Bush's "as we may think" (1945)

- Vannevar Bush's essay "As We May Think" is important in HCI courses and has garnered over 4,000 citations in scholarly publications.
- Bush was the U.S. government's Director of the Office of Scientific Research and a scientific advisor to President Franklin D. Roosevelt.
- Bush's essay concerned the dissemination, storage, and access to scholarly knowledge, and he was keenly aware of the possibilities of applying science to more lofty and humane pursuits.
- Bush proposed a device called memex, which had features like associative indexing, hyperlinks, and bookmarks.
- Bush's inspiration for memex came from the contemporary telephone exchange, which he described as a "spider web of metal."

- Some of Bush's ideas, like dry photography, seem naive today, but his vision of a world struggling with and fine-tuning and perfecting the dissemination and access to knowledge is still relevant.

1.3 Ivan Sutherland's Sketchpad (1962)

- Ivan Sutherland developed Sketchpad in the 1960s for his PhD research at MIT
- Sketchpad was a graphics system that allowed users to manipulate geometric shapes using a light pen
- Sketchpad was the first direct manipulation interface
- Direct manipulation interface is a suite of features, including visibility of objects, incremental action, rapid feedback, reversibility, exploration, syntactic correctness of all actions, and replacing language with action
- Dynabook concept system by Alan Kay of Xerox PARC followed the direct manipulation interface
- Sutherland's work on Sketchpad was presented at IEEE conference in 1963 and published in its proceedings
- Sketchpad demo videos are available on YouTube
- User testing of Sketchpad was not conducted as Sutherland was a student of electrical engineering

1.4 Invention of the mouse (1963)

- The computer mouse was invented by Douglas Engelbart in 1963.
- The mouse changed the way humans interact with computers by allowing users to manipulate a mouse to control an on-screen tracking symbol or cursor.
- The mouse replaced the light pen, which was awkward and fatiguing for users.
- The first prototype mouse had two potentiometers with large metal wheels attached to their shafts.
- The amount of rotation of each wheel altered the voltage at the wiper terminal of the potentiometer, which indirectly controlled the x and y positions of an on-screen object or cursor.
- The mouse was tested for selecting and manipulating text rather than drawing and manipulating graphic objects.
- An evaluation of the mouse was conducted in a controlled experiment comparing it to other input devices capable of selection and x-y position control of an on-screen cursor, including a light pen, a joystick, a knee-controlled lever, and a Grafacon.
- The knee-controlled lever and Grafacon were interesting alternatives to the mouse.
- Refining ideas is an iterative process that involves a good deal of trial and error.
- English et al. (1967) conducted a comparative evaluation of six input devices, including a mouse, measuring users' access time and motion time for each trial.
- The evaluation included 13 participants, and results were shown for mean task completion time and error rate.

- The knee-controlled lever appeared to be the best device in terms of time, but this was due to its zero access time, and considering motion time only, it did not show up favorably.
- The light pen had a slight advantage over the mouse in terms of time, but the mouse was the clear winner in terms of accuracy.
- The methodology of the study was empirical and included a detailed description of the participants, apparatus, and procedure.
- The evaluation included an independent variable, input method, with six levels, and two dependent variables, task completion time and error rate.
- Card et al. (1978) conducted a similar evaluation at Xerox PARC, which reconfirmed the advantage of the refined mouse over competing devices.
- Engelbart's invention of the mouse dates back to the early 1960s, but commercialization did not occur until 1981, when the Xerox Star was launched.
- Engelbart received the ACM Turing Award (1997) and the ACM SIGCHI Lifetime Achievement Award (1998).

1.5 Xerox star (1981)

- The National Computer Conference (NCC) was a yearly conference for computing
- The NCC was a gathering of researchers and a trade show with big and little players present
- Qupro Data Systems had a booth at the NCC
- The buzz at the NCC was about Xerox and their Star Information System
- The Xerox Star was the first commercially released computer system with a GUI
- The Star had windows, icons, menus, and a pointing device (WIMP) and supported direct manipulation and what-you-see-is-what-you-get (WYSIWYG) interaction
- The story of the Star began in 1970 when Xerox established its research center, PARC, in Palo Alto, California
- The Alto, the Star's predecessor, also had a GUI and mouse but was never released commercially
- The Star was an office automation system intended for business professionals to create, modify, and manage documents, graphics tables, presentations, etc.
- The Star workstations were connected via high-speed Ethernet cables and shared centralized resources, such as printers and file servers
- The Star used the desktop metaphor to bring concepts from the office desktop to the system's display
- The Star's icons represented things like documents, folders, trays, and accessories such as a calculator, printer, or notepad
- The Star's interactions dealt with files, not programs, making the system more usable and invisible to users.
- The Star interface was developed in the 1970s by a team at PARC led by Alan Kay.
- It was an example of direct manipulation, which allowed users to control a system through numerous input channels with a direct correspondence to a task.
- Direct manipulation systems were more flexible than previous command-line interfaces and allowed users to control the system according to their wishes.
- The Star was designed using a software paradigm known as event-driven programming, which was more complicated than sequential programming but allowed the system to respond asynchronously to user events.

- The Star was not a commercial success because it was not a personal computer but a networked workstation designed for an office environment.
- It was expensive and closed, meaning it could only run applications developed by Xerox.
- Other popular personal computer systems available around the same time were the Apple II and VisiCalc, the first spreadsheet application.

1.6 Birth of HCI (1983)

Nineteen eighty-three is a good year to peg as the birth of HCI. There are at least three key events as markers: the first ACM SIGCHI conference, the publication of Card, Moran, and Newell's The Psychology of Human-Computer Interaction (1983), and the arrival of the Apple Macintosh, pre-announced with flyers in December 1983. The Mac launch was in January 1984, but I'll include it here anyway.

1.6.1 First ACM SIGCHI conference (1983)

- Human-computer interaction's roots go back to 1969 with the formation of ACM's Special Interest Group on Social and Behavioral Computing (SIGSOC).
- SIGSOC initially focused on computers in the social sciences but soon shifted focus to the needs and behavioral characteristics of users, with talk about the user interface or human factors of computing.
- In 1982, SIGSOC became the ACM Special Interest Group on Computer-Human Interaction (SIGCHI).
- SIGCHI is an interdisciplinary group composed of computer scientists, software engineers, psychologists, interaction designers, graphic designers, sociologists, and anthropologists.
- The first SIGCHI conference was held in 1983 in Boston, and it has been held yearly ever since with an attendance of around 2,500 people in recent years.
- The technical program at CHI is competitive, with research papers peer-reviewed and acceptance requiring high-quality work.
- CHI papers have high visibility and impact in the field of HCI.
- SIGCHI sponsors or co-sponsors other conferences, such as the annual ACM Symposium on User Interface Software and Technology (UIST) and specialized conferences like the ACM Symposium on Eye Tracking Research and Applications (ETRA) and the ACM Conference on Computers and Accessibility (ASSETS).

1.6.2 The psychology of human-computer interaction (1983)

- "Card, Moran, and Newell" refers to the book "The Psychology of Human-Computer Interaction" published in 1983
- The book emerged from work done at Xerox PARC's Applied Information-Processing Psychology Project (AIP)
- The AIP mission was to create an applied psychology of human-computer interaction by conducting requisite basic research within a context of application

- The book contains 13 chapters organized roughly as scientific foundation, text editing examples, modeling, and extensions and generalizations
- The first 100 or so pages in the book provide a comprehensive overview of core knowledge in basic psychology as it pertains to the human sensory, cognitive, and motor systems
- Card, Moran, and Newell connected low-level human processes with the seemingly innocuous interactions humans have with computers by using the model human processor (MHP)
- The application selected to frame the analyses in the book was text editing
- Human behavior can be modeled as an information processing activity
- The Psychology of Human-Computer Interaction aimed to develop an engineering-style theory of the user that permitted approximate, back-of-the-envelope calculations of how the user would interact with the computer when operating at a terminal.
- Newell identified a dilemma in the field: experimentation cannot be done until it is too late
- Novel interactions in new products always seem to be followed by research papers identifying weaknesses and suggesting improvements
- The breakthroughs in products such as the iPhone, Wii, IntelliMouse, and Palm Pilot have context in the milieu of basic research in human-computer interaction and related fields
- Newell's objective in the book was to generate "tools for thought," including models of interaction that can be quantitative and predictive or qualitative and descriptive
- Newell suggests using approximate "back of the envelope" calculations to describe or predict user interaction
- Newell walks the reader through an interaction example using the model human processor to illustrate each step from stimulus to cognitive processing to motor response
- The example demonstrates the difficulty in accurately predicting human behavior
- The book has many other examples like this, and there are also modern contexts for the examples, just waiting to be found and applied

1.6.3 Launch of the Apple Macintosh (1984)

- January 22, 1984: Super Bowl XVIII took place
- Companies paid to advertise during the game to reach millions of viewers
- Apple aired a 60-second ad for the Macintosh personal computer during the third quarter
- The ad used Nineteen Eighty-Four as a theme, portraying the Mac as a computer that would shatter the conventional image of the home computer
- The ad was directed by Ridley Scott and was a hit, named the 1980s Commercial of the Decade by Advertising Age magazine
- Soon after the ad aired, computer enthusiasts purchased the Mac for its sleek design and latest input device, a computer mouse
- The Mac's operating system and applications software heralded the new age of the GUI with direct manipulation and point-select interaction

- The Mac's interface was simple and intuitive, with a one-button mouse that eliminated confusion on which button to press
- The history of Apple and the events leading to the release of the Mac are chronicled in various sources
- There is folklore surrounding Apple and its leaders, and some key events are listed in Figure 1.13, with the names of the key players deliberately omitted.

1.7 Growth of HCI and graphical user interfaces (GUIs)

- Formation of ACM SIGCHI in 1983 and success of Apple Macintosh in 1984 marked the beginning of human-computer interaction.
- Graphical User Interfaces (GUIs) became mainstream and a broader community of users and researchers were exposed to this new genre of interaction.
- Microsoft was a latecomer in GUIs, but gained market share with the release of Windows 3.0 (1990) and Windows 3.1 (1992) and improved versions of Windows like Windows 95, Windows 98, Windows XP, and Windows 7.
- Interest in HCI grew, and universities introduced courses in HCI or user interface (UI) design, with HCI emerging as a popular focus in other areas such as psychology, cognitive science, industrial engineering, information systems, and sociology.
- Companies realized the importance of good user interfaces and practitioners became important members of the HCI community.
- Bad UIs are a problem in HCI and practitioners work to apply HCI principles in industry.

1.8 Growth of HCI research

- Research interest in human-computer interaction initially focused on the quality, effectiveness, and efficiency of the interface.
- Empirical research in the early days of HCI focused on questions such as how quickly and accurately users can perform tasks with a GUI versus a text-based command-line interface.
- Menu design is a classic example of a research topic in HCI.
- Menus require recognition, and typing requires recall, making recognition preferred over recall in user interfaces, at least for novices.
- Organizing menu commands in a hierarchy that includes depth and breadth is one approach, but the best structure for the hierarchy is still a research question.
- Menu design issues were actively pursued as research topics in the early days of HCI, with depth versus breadth being one of the research issues.
- There are many other research issues in menu design, such as whether items should be ordered alphabetically or by function, whether the presence of a title on a sub-menu improves menu access, and whether auditory feedback improves menu access.
- Countless topics of research in HCI create opportunities for empirical research, and the answers to these research questions can be found in the papers cited.

CHAPTER 2

The Human Factor

- HCI faces challenges due to the complexity of humans
- Humans vary greatly in characteristics and abilities
- Designing systems that work for all humans is difficult and approximate
- HCI designers must "know thy user" to account for variability
- HCI researchers seek to understand human behavior and design better interactions
- Big questions like "why do humans make mistakes?" are difficult to answer empirically
- Empirical research requires narrow and well-defined questions and populations
- Descriptive models can help categorize and understand aspects of human behavior relevant to HCI

2.1 Time scale of human action

- Newell's Time Scale of Human Action is a descriptive model of the human
- The model has four bands, a biological band, a cognitive band, a rational band, and a social band.
- Each band is divided into three levels, and time is ordered by seconds and appears on a logarithmic scale.
- The most common dependent variable in experimental research in HCI is time.
- Newell's time-scale model is relevant to HCI as it reflects the multidisciplinary nature of the field.
- Research on selection techniques, menu design, force or auditory feedback, text entry, gestural input is within the cognitive band.
- Tasks related to web navigation, user search strategies, user-centered design, collaborative computing, ubiquitous computing, social navigation, and situated awareness occupy users for minutes or hours and are in the rational band.
- Tasks lasting days, weeks, or months are in the social band and might include workplace habits, groupware usage patterns, social networking, online dating, privacy, media spaces, user styles and preferences, design theory, etc.
- Research methods tend to be highly quantitative at the bottom of the scale and highly qualitative at the top of the scale.
- The transition between qualitative and quantitative methods moving from top to bottom in the figure is gradual.
- Researchers in the social band primarily use qualitative methods but often include some quantitative methods.
- Conversely, researchers in the cognitive band primarily use quantitative methods but typically include some qualitative methods.
- Newell speculates on bands above the social band: a historical band operating at the level of years to thousands of years, and an evolutionary band operating at the level of tens of thousands to millions of years.

2.2 Human factors

- There are multiple ways to characterize the human in interactive systems
- One way is the model human processor by Card et al. (1983)
- Human factors researchers often use a model of a human operator confronting a machine
- The model simplifies the human to three components: sensors, responders, and a brain
- The dashed vertical line represents the interface where interaction takes place
- This is where researchers observe and measure the behavioral events that form the interaction.

2.3 Sensors

- The five classical human senses are vision, hearing, taste, smell, and touch.
- Each sense brings different physical properties of the environment to the human.
- The senses convert physical phenomena into electrical nerve signals and transmit them to the brain for processing.
- Sensory stimuli and sense organs are purely physiological.
- Perception includes both the sensing of stimuli and use of the brain to develop identification, awareness, and understanding of what is being sensed.
- The section begins with a discussion of the first sense, vision.

2.3.1 Vision (Sight)

- Vision is the human ability to receive information from the environment in the form of visible light perceived by the eye.
- Most people obtain about 80% of their information through the sense of light.
- The act of seeing begins with the reception of light through the eye's lens.
- The retina at the back of the eye converts visible light into neurological signals sent to the brain via the optic nerve.
- The fovea, near the center of the retina, is responsible for sharp central vision and engages about 50% of the visual cortex in the brain.
- Light has properties such as frequency and intensity.
- Frequency is the property of light leading to the perception of color, and different colors are positioned within the visible spectrum of electromagnetic waves.
- Intensity is a more complicated concept that includes luminance and brightness.
- Vision engages in two primitive actions: fixations and saccades, during which the eyes take in visual detail from the environment and rapidly reposition to a new location.
- Eye movements have been analyzed in HCI research, including how people read and view content on web pages, which offers implications for page design.

2.3.2 Hearing (Audition)

- Hearing is the detection of sound by humans.
- Sound is transmitted through the environment as sound waves.
- Sound waves are created when physical objects are moved or vibrated.
- Hearing occurs when sound waves reach a human's ear and stimulate the ear drum to create nerve impulses that are sent to the brain.
- A single sound from a single source has at least four physical properties: intensity (loudness), frequency (pitch), timbre, and envelope.
- Loudness is quantified by sound pressure level, which expresses the pressure in a sound wave relative to the average pressure in the medium.
- Pitch is the subjective analog of frequency and the units of pitch are cycles per second or Hertz (Hz).
- Timbre (aka richness or brightness) results from the harmonic structure of sounds.
- Envelope is the way a note and its harmonics build up and transition in time.
- Sounds can be described as being harmonious or discordant.
- Humans have two ears, but each sound has a single source, and the slight difference in the physical properties of the sound as it arrives at each ear helps humans in identifying a sound's location.
- Sound is crucial for blind users, for example, in conveying information about the location and distance of environmental phenomena.

2.3.3 Touch (Tactition)

- Touch is just one component of the somatosensory system, which includes sensory receptors in the skin, muscles, bones, joints, and organs.
- The somatosensory system provides information on touch, temperature, pain, and body and limb position.
- Tactile feedback in HCI refers to information provided through the somatosensory system from a body part in contact with a physical object, including temperature, shape, texture, position, and resistance.
- All user interfaces that involve physical contact with the user's hands include tactile feedback.
- Tactile feedback is used in mouse and keyboard interactions to provide information on the feel of the mouse, the edges and shapes of keys, and resistance as a key is pressed.
- Tactile identifiers on key tops facilitate eyes-free touch typing.
- Augmenting the user experience through active tactile feedback is a common research topic.
- A solenoid-driven pin can be added to a mouse to provide tactile feedback when the mouse cursor crosses a boundary.
- Tactile feedback is commonly used in mobile phones through vibration to signal an incoming call or message.

2.3.4 Smell and taste

- Smell and taste are two senses that are less understood than the visual and auditory senses.
- Smell is the ability to perceive odors through sensory cells in the nasal cavity, while taste is a direct chemical reception of sweet, salty, bitter, and sour sensations through taste buds in the tongue and oral cavity.
- Flavor is a perceptual process in the brain that occurs through a partnering of the smell and taste senses.
- Complex smells and tastes can be built up from simpler elements, but the perceptual processes for this remain a topic of research.
- There are few examples in HCI where smell and taste are used, such as in aiding in searching digital photo albums or notifying users of an interruption by an incoming message.
- Empirical research methodologies have been used to explore the potential of smell in a user interface.
- In Brewster et al. (2006), users employed two tagging methods, text and smell, and then later used the tags to answer questions about the photos. Recall with smell tags was poorer than with word tags.
- In Bodnar et al. (2004), they compared smell, auditory, and visual modalities for notifying users of an interruption by an incoming message. They also found poorer performance with smell.

2.3.5 Other senses

- The term "sense" is used in various contexts apart from the five primary senses.
- Additional senses like a sense of urgency, musical sense, intuitive sense, moral sense, or common sense are valuable to HCI.
- These additional senses encapsulate how humans feel about their interactions with computers.
- Satisfaction, confidence, frustration, etc., are steeped in how users feel about computing experiences.
- Humans may have a moral sense similar to our sense of taste.
- It has been argued and supported with experimental evidence.
- Humans may have natural receptors that help us recognize fairness and cruelty.
- Just as a few universal tastes can grow into many different cuisines, a few moral senses can grow into different moral cultures.

2.4 Responders

- Movement or motor control empowers humans to affect their environment.
- Control occurs through responders.
- Movement involves the use of different body parts like fingers, feet, eyebrows, vocal chords, and torso.
- Penfield’s motor homunculus maps areas in the cerebral motor cortex to human responders, illustrating that muscles controlling the hand and fingers are highly

represented compared to other muscles responsible for the wrist, elbow, and shoulders.
- Card et al. (1991) hypothesized that muscles with a large area devoted to them are promising places to connect with input device transducers if high performance is desired.

2.4.1 Limbs

- Human control over machines is usually associated with upper body limbs
- Movement of limbs is coupled with proprioception for accuracy and finesse
- Limbs such as legs and feet can also act as responders in HCI
- Movement of head can control an on-screen cursor for users with limited arm mobility
- Handedness exists along a continuum and can be assessed with the Edinburgh Handedness Inventory
- Edinburgh Handedness Inventory is used in HCI experiments to determine participant handedness
- Occlusion of displays may affect performance or posture depending on handedness
- Positioning UI elements in a different region can avoid occlusion, but requires determining handedness of the user.

2.4.2 Voice

- The human vocal cords are used to create a variety of sounds
- Speech is the primary channel for human communication
- Speech can be recognized by algorithms implemented in software running on the host computer
- Non-verbal voice interaction (NVVI) is a modality that uses non-speech vocalized sounds to create an input channel
- NVVI is particularly useful for specifying analog parameters
- Examples of NVVI applications include controlling volume and simulating a joystick to control an on-screen cursor
- NVVI is primarily useful for creating accessible computing for users without a manual alternative.
- A system for mobile text entry using automatic speech recognition can achieve entry rates of 18 words per minute while seated and 13 words per minute while walking.

2.4.3 Eyes

- The human eye can act as a responder and control a computer through fixations and saccades
- Eye tracking apparatus is required to sense and digitize the gaze location and movement of the eyes for computer input control
- Eye tracking apparatus is configured to emulate a computer mouse for point-select operations
- Eye can look-select and activate soft controls such as buttons, icons, links, or text
- Fixating on a selectable target for a predetermined period of time can be used for eye selection

- Eye typing is an application of eye tracking for input control that uses an on-screen keyboard
- An iView X RED-III eye tracking device is an example of an eye tracking apparatus for computer input control
- A sequence of fixations and saccades (a scanpath) is used for eye typing and text entry
- Fixations are conscious and deliberate acts for controlling a computer interface, different from fixations used for viewing content on a web page.

2.5 The brain

- The brain is the most complex biological structure known
- The brain has billions of neurons
- The brain provides humans with a multitude of capacities and resources
- These capacities and resources include pondering, remembering, recalling, reasoning, deciding, and communicating
- The brain connects sensors (human inputs) and responders (human outputs)
- The brain's task begins upon experiencing the environment through sensors

2.5.1 Perception

- Perception is the first stage of processing in the brain where sensory signals are received as input from the environment.
- At the perceptual stage, associations and meanings take shape.
- Psychophysics examines the relationship between human perception and physical phenomena.
- Just noticeable difference (JND) is a common experimental goal in psychophysics to measure the difference between two stimuli that is not perceived by the subject.
- Ambiguity is the human ability to develop multiple interpretations of a sensory input.
- Illusion is the deception of common sense and can occur in visual, auditory, and tactile stimuli.
- The ability to go beyond the perceptual stage to excel at higher-level capabilities is what propels humans to the top tier in classification.

2.5.2 Cognition

- Cognition is the process of conscious intellectual activity such as thinking, reasoning, and deciding
- There are different views on the scope of cognition
- It is difficult to directly measure cognitive operations in the brain
- Sensory stimuli and motor responses can be measured to indirectly measure cognitive operations
- Techniques exist for measuring the duration of cognitive operations
- Examples of cognitive operations include responding to a changing signal while driving, pressing a button on a mobile phone, closing a popup ad while reading the news online, and switching to email while editing a document

- Some decisions, such as playing a card game, involve more complicated cognitive operations beyond the information in the sensory stimulus
- Memory is another major function of the brain.

2.5.3 Memory

- Memory is the human ability to store, retain, and recall information.
- Long-term memory is the brain's vast repository of experiences.
- Memory in the brain is organized similarly to memory in a computer with a declarative/explicit area for data and an implicit/procedural area for code.
- Working memory or short-term memory is an active area within long-term memory with a small capacity of about seven units.
- A study by Miller titled "The Magic Number Seven, Plus or Minus Two: Some Limits on our Capacity for Processing Information" found that humans can typically distinguish about seven levels of a uni-dimensional stimulus and can recall about seven items (±2) in a sequence.
- Chunking is a process in the brain where low-level items are grouped into a single high-level item, making it easier to remember.

2.6 Language

- Language is universally available to virtually all humans
- Language as speech is available without conscious effort and children learn it as they grow and develop
- Writing is a more recent phenomenon and demands considerable effort and years of study and practice
- Civilization is defined by writing, which enables the cultural and technological status associated with it
- Prehistory dates from the arrival of human-like beings to the emergence of writing
- In HCI, the interest in language is primarily in systems of writing and technology that enable communication in a written form
- A corpus is a large collection of text samples gathered from diverse and representative sources
- Corpora are used to study a language in its written form
- The sampling process of a corpus brings limitations, such as the timeframe, country, region, topic, and writer
- The British National Corpus (BNC) is a well-known corpus that includes samples totaling 100 million words written in British English from the late 20th century
- A corpus is sometimes reduced to a word-frequency list, which tabulates unique words and their frequencies in the corpus
- The most frequent word in the BNC is "the", representing about 6.8 percent of all words
- Figure 2.19 includes excerpts from several corpora showing the five most frequently used words and words ranked from 1000 to 1004
- There are additional columns for French, German, Finnish, SMS English, and SMS Pinyin

- SMS text messaging is a good example of the dynamic and context-sensitive nature of language
- Pinyin is the standard coding system using the Latin alphabet for Mandarin Chinese characters
- Part-of-speech (POS) tagging is where words are tagged by their category, such as noun, verb, and adjective, and can be important in predictive systems

2.6.1 Redundancy in language

- Native speakers possess innate understanding of language statistics
- Humans can fill in missing letters or words
- Text can be shortened without losing comprehension
- SMS messaging uses recoding techniques like sound and acronyms
- SMS shorthand can drastically reduce character count
- Redundancy in written text can be detrimental
- Superfluous words should be omitted in writing style

2.6.2 Entropy in language

- Redundancy and entropy are related concepts in language
- Shannon conducted a letter-guessing experiment in the 1950s to demonstrate redundancy and entropy in written English
- The experiment showed that errors in guessing letters are more common at the beginning of words, less common as words progress
- Shannon quantified the entropy of English in information-theoretic terms, showing that the reduced text contains the same information as the original text with a good statistical model
- The entropy of printed English is about 4.25 bits per letter considering letter frequencies alone
- Considering previous letters and long-range statistical effects, the entropy of printed English is reduced to about one bit per letter with a corresponding redundancy of about 75 percent.

2.7 Human performance

- Humans use sensors, brain, and responders to perform tasks, leading to human performance
- Human performance is associated with the speed-accuracy trade-off: going faster leads to errors while slowing down improves accuracy
- Research on new interfaces or interaction techniques must consider both speed and accuracy
- Humans position themselves on the speed-accuracy trade-off based on their comfort level and goals, and context plays an important role
- HCI faces complexities and challenges due to human diversity, variability, and imprecision

- Human proficiency in tasks varies and can be illustrated in a distribution
- Typing speeds on a conventional keyboard range from 30-70 words per minute, with some people being slower or faster and a small number being exceedingly fast or slow

2.7.1 Reaction time

- Simple reaction time is a primitive manifestation of human performance, involving the delay between a stimulus and a response.
- Simple reaction time involves a cognitive operation that is trivial, making it an easy task to study.
- There are many stimuli that can elicit a simple reaction time response, such as auditory, visual, and tactile stimuli.
- Reaction times vary according to the type of stimulus, with auditory being the fastest and pain being the slowest.
- A Java-based application was developed to experimentally test several reaction time tasks that add complexity to the cognitive operation.
- The application includes three extensions to simple reaction tasks: physical matching, name matching, and class matching.
- Physical matching involves comparing two stimuli to determine if they match.
- Name matching is similar to physical matching, but the stimuli vary in appearance.
- Class matching involves determining if two stimuli belong to the same class, either letters or digits.
- In a course on HCI, fourteen students performed reaction time tasks and the results were analyzed.
- Simple reaction took an average of 276ms.
- Physical matching took about twice as long as simple reaction.

2.7.2 Visual search

- Visual search is a variation on reaction time, where the user scans a collection of items to find a desired item
- The time required for visual search increases with the number of items to scan
- The software includes a mode for visual search with the search space configurable for 1, 2, 4, 8, 16, or 32 items
- A small experiment was conducted using the same 14 students from a previous experiment, and a linear relationship was found between reaction time in a visual search task and the number of items to scan
- The results are consistent with previous research on menu selection
- The experiment revealed that RT = 498 + 41 x Nms
- The no-match trials take longer than the match trials because an exhaustive search is required to determine if the initial stimulus is not present
- False start in athletics is declared if an athlete reacts within 100ms after the starter's pistol is sounded or before it is sounded

- The lower-bound reaction times cited above are for pressing a key with a finger in response to a visual stimulus, while the motor response signals in the 100 meter dash must travel farther to reach the feet.

2.7.3 Skilled behavior

- Response time tasks are simple and involve a sensory stimulus, cognitive operation, and motor response
- Human performance improves considerably and continuously with practice in many tasks
- Skilled behavior is a property of human behavior whereby human performance necessarily improves through practice
- Examples of skilled behavior include playing darts, playing chess, gaming, and programming
- Skilled behavior can be categorized as sensory-motor skill or mental skill, but all skilled behavior requires mental faculties and coordinated action
- Tasks such as minimally invasive surgery involve both sensory-motor and mental skill
- Studying skilled behavior involves recording and charting the progression of skill over time using dependent variables such as speed and accuracy, and temporal units such as minutes, hours, days, months, or years
- Methodology for evaluating skilled behavior is presented in Chapter 5 with mathematical steps for modeling presented in Chapter 7.

2.7.4 Attention

- Texting while driving is dangerous and can be deadly
- Human ability to attend to tasks is limited
- Attention is a property of human behavior that occurs when a person who is attending to one thing cannot attend to another
- Typing requires attention because while typing we cannot engage in conversation
- Walking requires very little attention since we can think, converse, and do other things while walking
- Attention is often studied along two themes: divided attention and selected attention
- Divided attention is the process of concentrating on and doing more than one task at a time
- Selected attention is attending to one task to the exclusion of others
- Selective attention is the human ability to ignore extraneous events and to maintain focus on a primary task
- Importance of events is subjective and can affect selective attention
- The distinction between divided and selected attention is often explained in terms of channels
- Analyzing accidents is an important theme in human factors
- Accidents are often caused by distractions or selectively attending to inappropriate events
- The statistics of accidents caused by texting while driving are shocking

- Attention has relevance in HCI in office environments where interruptions affect productivity
- The mobile age has brought a milieu of issues bearing on attention, including immediate, brief tasks that demand constant vigilance and user availability.

2.7.5 Human error

- Human error is important in examining performance in HCI experiments
- Errors are discrete events in a task where the outcome is incorrect
- Errors are reported as the ratio of incorrectly completed trials to all trials, often as a percent
- Accuracy is reported as the ratio of correctly completed trials to all trials
- Examples of computing tasks with errors are shown in Figure 2.32
- Categorizing tasks as correct or incorrect does not fully capture behavior
- Human error is often studied to understand how and why errors occur
- Environmental circumstances can be a factor in errors
- Human factors researchers examine human error in industrial accidents
- Failures in industrial accidents are systemic, often due to design-induced errors
- Interaction errors by an operator must be considered and accounted for in the design of safety-critical systems

CHAPTER 3

Interaction Elements

- Human-computer interaction involves a human performing a task using computing technology, which can be goal-oriented or not.
- The text examines low-level aspects of interaction in the cognitive band of Newell's Time Scale of Human Action.
- Most experimental research in HCI is directed at tasks with durations ranging from 100ms to 10s, as extraneous behaviors are easy to control.
- Human input to computers and the responses produced are discussed, along with the importance of input and output in the human factors view of interaction.
- The text notes the history of human factors research and how it remains relevant in newer domains such as mobile computing, virtual environments, and gaming.

3.1 Hard controls and soft controls

- Before computers, displays and controls were physical, single-purpose devices that were fixed for life.
- The advent of computers and software brought about soft interfaces, which changed the way humans interact with technology.
- Soft controls can be created quickly through software and bring unlimited possibilities to a small physical space.
- Human-computer interfaces contain lots of controls, most of which are soft controls manipulated by physical controls like the mouse or keyboard.
- Soft controls blur the distinction between controls and displays since they are rendered on a display through software and can dynamically change size, shape, or appearance.
- Most soft controls are also displays and can convey information to the user.
- Soft controls are manipulated by hard controls like clicking, double-clicking, dragging, or keyboard access.
- The hard-control/soft-control distinction is similar to the idea of space multiplexing versus time multiplexing.
- In the next section, the book examines controls and displays in terms of their spatial relationships, starting with hard controls like a mouse and typical displays found on desktop computer systems.

3.2 Control-display relationships

- The text discusses the control-display relationship in human-computer interaction, where a user's action results in a corresponding response on the system's display.
- The example of moving a computer mouse to the right and having the cursor move to the right is used to illustrate this relationship.
- The control-display relationship is described as natural, seamless, intuitive, and efficient.
- Control-display relationships are also referred to as mappings, which attribute how a controller property maps to a display property.

- The text mentions that there are different types of relationships in human-computer interaction, including spatial relationships, dynamic relationships, and physical relationships.
- Spatial relationships are described as relating to the position or movement of a controller and its corresponding effect on the display.
- The text implies that the following sections will discuss dynamic and physical relationships in more detail.

3.2.1 Spatial relationships

- The mouse-cursor spatial relationship is illustrated in Figure 3.4a.
- The mapping is congruent because there is an exact spatial correspondence between the controller input and the display output.
- Figure 3.4b shows a different situation where move the mouse “forward” and the cursor moves “up.”
- The spatial relationship between mouse and cursor is learned very quickly.
- Figure 3.5a and Figure 3.5b show the display and mouse pad surfaces in a Cartesian coordinate space with x-, y-, and z-axes.
- Mouse x-axis motion maps to cursor x-axis motion, but mouse z-axis motion maps to cursor y-axis motion.
- Aiming error is known to be higher for 90° to 135° transformations and lower for 0° or 180° transformations.
- Adaptation occurs for all transformations, in a few as 50 trials.
- Scrolling involves manipulating a physical controller to move a pointer to the slider and acquiring it with a button-down action.
- There is a spatial transformation between the hard control and soft control and between the soft control and the view on the display.
- The +/− reversal in the control-display relationship along the z-axis is interesting.
- Six parameters are required to fully place an object in 3D space: three for the object’s position in space (x, y, z), and three for the object’s orientation in space (θx, θy, θz).
- In aeronautics, the rotation terms are known as pitch (θx), roll (θz), and yaw (θy).
- Figure 3.8a shows an example of 3D interaction where congruence is fully attained

3.2.2 CD gain and transfer function

- CD gain is a property of mouse-cursor movement that represents the amount of movement in a display object for a given amount of movement in a control.
- CD gain is typically set in a control panel, and lowering it reduces the amount of controller movement required for each unit of cursor movement.
- The relationship between controller motion and cursor motion is often non-linear and uses a power function, which is enabled by selecting "Enhance pointer precision."
- Research on optimizing CD gain dates back to at least the 1940s, and has been highly researched over the past 20 to 30 years as it applies to interaction with GUIs on desktop computing systems.
- The majority of studies have found that significant performance benefits are not achieved by adjusting CD gain.

- One challenge in optimizing CD gain is defining optimal performance, as the goal of optimizing both speed and accuracy is problematic due to the speed-accuracy trade-off.
- Optimizing CD gain is of continuing interest today in different settings, such as very large displays, very small displays, accessible computing, remote pointing, and 3D interaction.
- Since human performance is the main criterion for optimization, research on CD gain is well-suited to empirical enquiry and experimentation.

3.2.3 Latency

- Human interaction with computers is a two-way structure where we issue commands through input controllers and receive responses on output displays.
- Latency, the delay between an input action and the corresponding response on a display, can adversely affect human performance and the interaction experience when feedback is delayed.
- Latency is negligible in many interactive computing tasks, such as typing or cursor positioning.
- Remote manipulation, such as controlling a space vehicle on a distant moon or planet, has significant latency due to mechanical linkages or transmission delays in the communications channel.
- Latency is often present with internet connections, and frustration can be reduced by including feedback informing the user that the network is processing the request.
- Virtual reality relies heavily on the tracking of hand, head, or body motion in a simulated 3D environment, and the pretense of reality requires a tight link between the user's view of the environment and the actions that set the view. When changes in the environment lag behind input motions, the loss of fidelity is dramatic.
- Latency is attributable to properties of input devices, output devices, and software, and latency increases due to software overhead and other system-related factors such as network configurations, number crunching, and application software.
- A proper latency measurement is done externally, as it is not possible to accurately measure latency from within the target system.
- Latency in virtual reality is associated mostly with 6 DOF tracking devices attached to the hand, head, or body, as well as the latency caused by low frame rates.
- Low latency is significantly more important than high-quality graphics or stereoscope in VR systems.
- Empirical measurements of the effect of latency on human performance are rare

3.2.4 Property sensed and order of control

- Input controllers involve actions such as touching, tapping, grasping, moving, pushing, flicking, and squeezing, which are sensed and converted into data for processing by the host computer.
- Pointing devices sense properties such as position, displacement, and force.
- Position is sensed as an absolute coordinate at the point of contact along the x-axis and z-axis for graphics tablets, touchpads, or touchscreens.

- Mouse displacement is sensed as the amount of movement along the x-axis and z-axis relative to the last sample.
- The most common orders of control are position-control and velocity-control.
- Joysticks are either isotonic or isometric.
- Isotonic joysticks are also called displacement joysticks, and the output signal represents the position of the stick as the amount of displacement from the neutral or home position.
- Isometric joysticks sense the force applied to the stick and the output signal represents the amount of force along the x-axis and z-axis.
- Kantowitz and Elvers evaluated an isometric joystick in both position-control and velocity-control modes and found velocity-control to perform best.
- Zhai conducted an experiment and observed that position control is best for a position-sensing device and that velocity control is best for a force-sensing device.

3.3 Natural versus learned relationships

- The article discusses the importance of a control-display relationship and how it can affect usability.
- The naturalness of the relationship between control motion and display response is examined, with an example in Figure 3.15.
- The relationship is not natural but learned, even if it appears simple and intuitive.
- Warrick's principle is discussed as an example of applied ergonomics, but it has little basis in human sensory-perceptual cognition.
- The best possible scenario is spatial congruence, which implies using a sliding or linear control.
- The article discusses how population stereotypes can affect the naturalness of the relationship between control and display, even in cases where there is no physical analogy.
- The importance of a physical mapping that is compatible between the stimulus (control) and response (display) is highlighted, with the example of elevator buttons.
- Physical misalignment between displays and controls or between controls and their responses can lead to errors and longer response times.
- The article concludes by emphasizing the importance of this work for HCI.

3.4 Mental models and metaphor

- Physical analogy and metaphor are common ways to learn and adapt.
- Mental models, also known as conceptual models, are common in Human-Computer Interaction (HCI).
- HCI’s first mental model was the office or desktop metaphor that helped users understand the graphical user interface.
- Implementation models, which impose a set of interactions that follow the inner workings of an application, should be avoided in designing software-based products.

- Toolbars in GUIs use icons as pictorial representations to trigger a mental image in the user's mind associated with the action of the button or tool.
- Hovering the mouse pointer over a GUI button in Apple's method pops up a field providing a terse elaboration on the button's purpose.
- Compass and clock face are examples of mental models for direction in HCI.
- Users wore a vibro-tactile belt with eight actuators positioned according to compass directions to navigate the virtual building using a mental model of the compass.

3.5 Modes

- Modes are a common and frustrating property of user interfaces that occur when there are fewer controls than tasks.
- A mode is "a functioning arrangement or condition."
- Systems of all types have modes, including office phones, electric staplers, lawn mowers, and traffic intersections.
- Challenges with modes occur in interactive systems, such as keyboards, because they have fewer controls than tasks.
- Function keys and some physical keys on mobile phones and LCD computer monitors have many modes.
- The Navi key on Nokia 3210 was the first physical key with modes, and it used the display to show the current mode of the key.
- Soft keys, like those in Figure 3.3, have an infinite number of modes.
- Designing modeless interactions is sometimes desirable but not always possible since modes are unavoidable in most cases.
- The text editor vi is a pre-GUI legacy application used by expert computer users.
- GUI-based systems require a higher standard of usability than vi due to their broad user base.
- Insert versus Overtype mode in Microsoft Word is an example of a system with mode problems, including changing modes and mode visibility.
- Mode visibility in Microsoft Word is inadequate, leading to a gulf of evaluation between the system and the user.
- Feedback is crucial to improving user interaction with modes, and audio and visual feedback could be explored as solutions.
- Modes are common in graphics and 3D interaction, with control-display operations often using rotate mode.
- A rotate mode presents a disconnect between the capabilities of 2D pointing devices, such as a mouse, and the characteristics of a 2D surface.
- The use of a rotate handle in newer versions of PowerPoint eliminates the need for a rotate mode, but control-display relationships remain the same.
- Mouse reengineering can sense rotation and eliminate the need for a rotate mode, with several possible configurations available.

3.6 More about degrees of freedom

- HCI research literature has numerous examples of devices engineered to provide additional sensing capabilities.
- Rockin'Mouse is a 4 DOF mouse that senses planar x-z movement and also senses rocking side-to-side and to-and-fro rotation.
- The three-axis trackball by Evans et al. (1981) senses rotation about each of the x-, y-, and z-axes.
- Jacob et al. (1994) speak of the integrality and separability of the degrees of freedom. Degrees of freedom can be separated or integrated.
- The Etch-A-Sketch children's toy introduced in the 1950s by the Ohio Art Company is an example of a device with separate 1 DOF controllers to move the drawing instrument along the x- and y-axes.
- The wheel mouse, released in 1996 as Microsoft's IntelliMouse, has an extra—and separate—degree of freedom.
- Balakrishnan and Patel's PadMouse (1998) combines a touchpad with a regular mouse to support pie menu access with gestures on the touchpad while the mouse operates as usual, controlling an on-screen cursor.
- Silfverberg et al. (2003) describe a display screen that allows the displayed image to be zoomed and panned by two touch sensors positioned underneath the device.
- Villar et al. (2009) describe a variety of mouse configurations instrumented to detect touch, including multi-touch.

3.7 Mobile context

- Mobile computing started in the 1980s with pocket organizers and PDAs, followed by the first pen-based tablet computers and mobile phones supporting text messaging in the early 1990s.
- The release of the Apple iPhone in 2007 was a significant event for mobile computing as it led to a dramatic shift in the landscape of human-computer interaction (HCI) with finger actions like swiping, flicking, pinching, tapping, and more.
- Smart devices like smartphones and tablets include components like cameras, light sensors, vibro-tactile actuators, accelerometers, gyroscopes, GPS receivers, and more to enhance the user experience and provide unprecedented connectivity to users through both 3G and internet access.
- Direct input through touchscreens and indirect input through cursor control on desktop environments are different, with the former being more WYSIWYG and not requiring an on-screen tracking symbol like a cursor or pointer.
- Direct touch input has the "fat finger problem," making precise pixel-level selection difficult because the finger occludes the location of contact, and finger contact forms an area, not a point.
- Empirical research is needed to solve these challenges, and one of the earliest selection enhancements for touchscreens was the offset cursor presented by Potter et al. (1988) at the ACM SIGCHI conference in 1988.
- Contemporary touchscreen devices like the iPhone's soft keyboard use variations of Potter et al.'s offset cursor combined with take-off selection.

- Touch-sensing technologies have the ability to sense multiple contact points, and multi-touch is one of the most provocative interaction techniques to enter mainstream computing in recent years

3.8 Interaction errors

- The chapter discusses physical properties of human-machine interface (HCI) such as degrees of freedom and spatial relationships.
- Human performance is important, but it is not discussed in this chapter except for secondary observations.
- Empirical research in HCI is largely about finding physical properties that enhance human performance.
- Interaction errors frustrate users and hinder performance.
- The difficult problems in desktop computing are partially solved, but many frustrating UI problems persist.
- Four examples of UI problems are presented, ranging in severity from serious to innocuous.
- One serious problem is the loss of information due to mistaken discarding, overwriting, or loss of data.
- Another interaction error is logging in with caps_lock mode in effect.
- The chapter suggests that the big problems get attention and get fixed, but small problems tend to linger.
- When an application or dialog box is active, one of the UI components has focus and receives an event from the keyboard if a key is pressed.
- For buttons, focus is usually indicated with a dashed border. For input fields, it is usually indicated with a flashing insertion bar ("|").
- There is wide-spread inconsistency in current applications in the way UI widgets acquire and lose focus and in the way focus advances from one component to the next, making the user uncertain about where the focus is.
- Many online activities require users to enter data into a form, but users often don't know if focus will automatically advance to the next field or if a user action is required.
- Data entry fields have not reached the evolutionary status of, for example, dialog boxes for saving versus discarding changes.
- The absence of expectations keeps the user on guard, resulting in a slight increase in the attention demanded during interaction, which produces a slight decrease in transparency.
- Each time a user shifts his or her attention (e.g., from the keyboard to the display and back), the cost is two gaze shifts, and these little bits of interaction add up.

CHAPTER 4

Scientific Foundations

- Chapter discussed various interaction topics in HCI
- Research methodology for studying these topics is empirical and scientific
- Ideas are conceived, developed, and implemented and framed as hypotheses
- Hypotheses are tested in experiments
- Chapter presents enabling features of this methodology
- Goal is to establish what, why, and how of research, with focus on empirical and experimental research
- Discussion is general, but examples are directed at HCI
- Chapter begins with terminology surrounding research and empirical research.

4.1 What is research?

- Research has different meanings for different people.
- The term "research" is often used simply to add weight to an assertion.
- Research is more than just a word to add force to a statement or opinion.
- Research has at least three definitions: conducting research can be a simple exercise of careful or diligent search, collecting information about a particular subject, and investigation or experimentation aimed at the discovery and interpretation of facts and revision of accepted theories or laws in light of new facts.
- Experimentation is a central activity in a lot of HCI research.
- HCI research may have non-experimental methodologies, and the book focuses on empirical research, which encompasses both experimental and non-experimental methodologies.
- Facts are the building blocks of evidence, and evidence is sought in experimental research.
- Theories and laws are mentioned in the third definition of research.
- A law is more specific, more constraining, more formal, more binding than a theory.
- Laws are of questionable relevance to HCI.
- Fitts' law refers to a body of work that pertained to rapid-aimed movements and is widely used in HCI.

4.1.1 Research must be published

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- Publication is the final and essential step in research.
- "Publish or perish" is an important edict for researchers, especially in academia.
- Weak or insufficient publication lists can hinder the chances of getting research funds or tenure-track professorship.
- Skill in publishing starts as a graduate student and continues throughout a researcher's career.
- Published research is critical for extending, refining, or revising the existing body of knowledge in the field.

- Publication in archived peer-reviewed journals or conference proceedings is necessary for high-standard scrutiny.
- Peers review the research for its integrity, relevance, and contribution.
- Archived work is added to the collection of related work accessible to researchers worldwide.
- Patenting is an option for inventors or companies wishing to profit from their inventions.
- Patenting involves disclosing the invention in a patent application that also describes prior art, how the invention addresses a need, and the best mode of implementation.
- Patents are also a form of publication that meet the must-publish criterion for research.

4.1.2 Citations, references, impact

- Hyperlinks connect web pages to other web pages, providing structure and cohesion to a topic or theme.
- Citations connect research papers to other research papers, shaping a body of research and guiding later research.
- Citations serve the purpose of supporting intellectual honesty, acknowledging previous work, and backing up assertions.
- The number of citations to a research paper is a measure of its impact and significance to the field.
- The H-index is the most accepted single measure of the impact of a researcher's publication record, quantifying research productivity and overall impact of a body of work.
- The H-index was proposed by physicist J. Hirsch in 2005.
- The formatting of citations and references is discussed in Chapter 8.
- Before services like Google Scholar emerged, citation counts were difficult to obtain.
- Citation counts are available for individual papers and easy to compile for individual researchers.

4.1.3 Research must be reproducible

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- Replicability is crucial in research
- Standardized methodology enforces a process for conducting and writing about research that ensures sufficient detail is included to allow results to be replicated
- Many great advances in science and research pertain to methodology
- Louis Pasteur's contribution was using a consistent methodology for his research in microbiology
- Pasteur's experiments and findings were controversial at the time, and the best way to fend off skepticism was to empower critics by adopting a methodology that allowed his experiments and findings to be replicated
- Reviewers of manuscripts submitted for publication often critique the work on the replicability point

- Lowry et al.'s 1951 paper "Protein Measurement With the Folin Phenol Reagent" is one of the most cited papers in publishing history and describes a method for measuring proteins in fluids

4.1.4 Research versus engineering versus design

- Engineers and designers build products that aim to balance form (design emphasis) and function (engineering emphasis).
- Research is focused on small ideas that are prototyped, tested, and then either advanced or discarded, leading to incremental progress.
- Researchers work with prototypes to inform or assess ideas at an early stage, unlike designers and engineers who use them at a relatively late stage of product development.
- Research precedes engineering and design and has a slower pace without the pressure of deadlines.
- The timeline for research, engineering, and design is shown in Figure 4.3, with products being the focus of deadlines.
- The success of products like the computer mouse and iPhone is due in part to the raw materials that came from research conducted before the engineers and designers began their work.
- Apple's engineers and designers were guided or inspired by research that came before them, such as multi-touch gestures and tilt as an interaction technique for user interfaces.

4.2 What is empirical research?

- Empirical research adds powerful new ideas to research.
- Empirical means based on observation or experience, or relying on experience or observation alone without due regard for system and theory.
- Researchers should be guided by direct observations and experiences without prejudice to existing theories.
- Nicolas Copernicus found evidence to the contrary of the prevailing system that celestial bodies revolved around the earth through empirical evidence based on observation without bias.
- Empirical evidence can lead to great achievements in modern science.
- In HCI and other fields of research, discoveries are usually more modest.
- Empirical means capable of being verified or disproved by observation or experiment.
- An HCI research initiative is framed by hypotheses that must be sufficiently clear and narrow to enable verification or disproval by gathering and testing evidence.
- Language used in an assertion should speak directly to empirical, observable, quantifiable aspects of the interaction.

4.3 Research methods

- Three common research methods: observational, experimental, and correlational
- All three methods are empirical, based on observation or experience
- Differences in methods depend on research objectives and researcher's expertise and style.

4.3.1 Observational method

- Observational, experimental, and correlational methods are common approaches for conducting research in HCI and other natural and social sciences.
- The observational method is based on observation and starts with observing humans interacting with computers or computer-embedded technology.
- The observational method includes techniques such as interviews, field investigations, case studies, focus groups, and more.
- The approach tends to be qualitative rather than quantitative, achieving relevance while sacrificing precision.
- Observational methods aim to explain the reasons underlying human behavior and focus on human thought, feeling, attitude, emotion, passion, sensation, reflection, expression, sentiment, opinion, mood, outlook, manner, style, approach, and strategy.
- Observations are more likely to involve note-taking, photographs, videos, or audio recordings rather than measurement.
- Observational methods tend to examine and record the quality of interaction rather than quantifiable human performance.

4.3.2 Experimental method

- Knowledge is acquired through controlled experiments conducted in laboratory settings with the experimental method
- Acquiring knowledge may imply gathering new knowledge or studying existing knowledge for various purposes
- The relevance-precision dichotomy explains the controlled experiments' diminished relevance but increased precision due to the reduced extraneous factors
- A controlled experiment in HCI requires at least two variables: a manipulated variable and a response variable
- A manipulated variable is typically a property of an interface or interaction technique presented to participants in different configurations
- A response variable is a property of human behavior that is observable, quantifiable, and measurable
- The methodology employed in HCI experiments is borrowed from experimental psychology
- A full and proper user study involves more than just measuring and analyzing human performance
- Qualities of interaction outside the scope of the experimental procedure cannot be ignored
- Cause-and-effect conclusions are possible in research using the experimental method if designed and conducted properly, but not in observational or correlational methods

4.3.3 Correlational method

- Correlational method involves looking for relationships between variables
- Researcher might be interested in knowing if users' privacy settings in a social networking application are related to their personality, IQ, level of education, employment status, age, gender, income, etc.
- Data is collected on each item and relationships are examined
- The magnitude of variables must be ascertained for correlational methods
- Correlational methods often accompany experimental methods
- Correlational methods provide a balance between relevance and precision
- Data collected using informal techniques bring relevance to real-life experiences
- Data obtained using correlational methods are circumstantial, not causal
- The book is primarily directed at the experimental method for HCI research but it is clear that the experimental method will often include observational methods and correlational methods.

4.4 Observe and measure

4.4.1 Observation

- The starting point for empirical research in HCI is observing humans interacting with computers
- Observations can be made by a human or an apparatus
- Human observation involves manually collecting measurements, which could involve using a log sheet or notebook
- Events of interest might include the number of times a user clicked a button or moved their hand from the keyboard to the mouse
- Manual observation could also involve timing by hand the duration of activities
- Observations may be broken down by gender or some other attribute of interest
- More often, observing is delegated to the apparatus
- The challenge in some situations is embedding data collection software in the apparatus
- Some behaviors of interest are difficult to detect, and additional apparatus or sensing technology may be required
- Custom software implementing an interface or interaction technique usually makes it straightforward to record events such as key presses, mouse movement, selections, finger touches, or finger swipes

4.4.2 Measurement scales

- Observation alone is limited and measurement is necessary for serious examination of patterns.
- Measurement involves observing, quantifying, and recording data.
- Lord Kelvin emphasized the importance of measurement for knowledge.
- There are four scales of measurement: nominal, ordinal, interval, and ratio.
- These scales determine the type of information and analyses possible in research.

4.4.3 Nominal

- Nominal scale involves arbitrarily assigning a code to an attribute or category
- Mathematical manipulations on nominal data are meaningless
- Nominal data identify mutually exclusive categories, and the only relationship that holds is equivalence
- Nominal data are also called categorical data
- Nominal data are often used with frequencies or counts
- Counts are a ratio-scale measurement
- Example of nominal data using real data: students were observed and categorized by gender and whether they were using a mobile phone while walking
- Research question: are males or females more likely to use a mobile phone while walking on a university campus?

4.4.4 Ordinal data

- Ordinal scale measurements provide an order or ranking to an attribute
- The attribute can be any characteristic or circumstance of interest
- Ordinal data is limited because the interval is not intrinsically equal between successive points on the scale
- Examples of ordinal data include ranking preferences of GPS systems and ordering features of a mobile phone by personal importance
- It is possible to compare greater than or less than with ordinal data
- It is not valid to compute the mean of ordinal data
- A questionnaire can be used to collect ordinal data, such as ranking the number of e-mail messages received per day

4.4.5 Interval data

- Interval data have equal distances between adjacent values, but there is no absolute zero.
- Mean can be computed for interval data, but ratios are not meaningful.
- Likert Scale is an example of interval data commonly used in questionnaires.
- Verbal responses are given a numeric code, and responses are symmetric about a neutral central value.
- There is disagreement among researchers on the assumption of equal gradations between responses.
- Respondents perceive items at the extremes of the scale as farther apart than items in the center.
- One remedy for non-equal gradations in Likert-scale response items is to instruct respondents to interpret the items as equally spaced.
- Likert Scale questionnaire items are commonly used in HCI research papers.

4.4.6 Ratio data

- Ratio-scale measurements are the most advanced of the four scales of measurement
- Ratio data have an absolute zero and support a wide range of calculations
- In HCI, the most common ratio-scale measurement is time, but physical and social variables are also measured using the ratio scale
- Count is another common ratio-scale measurement used in HCI research, where the number of occurrences of certain human activities are counted
- Normalizing counts improves their expressive nature and facilitates comparisons
- An experiment comparing five different text entry methods reported results in "words per minute" for speed, but in terms of the number of errors committed for accuracy
- Results should be normalized to make them more meaningful and facilitate comparisons, such as reporting error rates (%), computed as the number of character errors divided by the total number of characters entered times 100.

4.5 Research questions

- In HCI, experimental research is conducted to answer questions about user interface or interaction technique
- Research questions often pertain to the relationship between two variables
- Variability in human actions affects the confidence with which research questions can be answered
- Statistical techniques are used to gauge the confidence of answers
- Research questions emerge from an inquisitive process
- Initial thoughts are fluid and informal
- The goal is to move forward to questions more suitable for empirical and experimental enquiry
- An example is given of a researcher interested in text entry on touchscreen phones
- The researcher builds a prototype of a new entry technique and decides to undertake experimental research to evaluate it
- Research questions become progressively narrower and more focused
- Expressions like "any good" or "better than" are problematic for research
- The last question is a specific, testable research question that addresses qualities that are easily observed and measured.

4.6 Internal validity and external validity

- Experimental research has two important properties: internal validity and external validity.
- Internal validity measures the extent to which an observed effect is due to the test conditions.
- External validity measures the extent to which experimental results are generalizable to other people and other situations.
- High internal validity means the effect observed really exists.
- Generalizable research results apply to other people and other situations.
- Experiment design is an exercise in compromise.

- The tension between internal and external validity is characterized in Figure 4.9.
- Environmental and procedural considerations can negatively impact internal validity, even if they improve external validity.
- Random sampling is important for generalizability in experimental research.

4.7 Comparative evaluations

- Evaluating new ideas for user interfaces or interaction techniques is important in human-computer interaction research.
- Some evaluations in HCI focus on a single idea or interface and do not include comparisons.
- Comparative evaluation is more appealing from a research perspective as it involves experimentation, discovery, and developing theories of interaction.
- In a comparative evaluation, a new user interface or interaction technique is designed and implemented and then compared with one or more alternative designs.
- A controlled experiment must include at least one independent variable and the independent variable must have at least two levels or test conditions.
- Including an established design as a baseline condition is appealing as it serves as a check on the methodology and allows results to be compared with other studies.
- Comparative evaluation yields more valuable and insightful results than a single-interface evaluation.
- Participants interacting with all three interfaces consistently found more problems and were more critical of the interfaces in a comparative evaluation.

4.8 Relationships: circumstantial and causal

- HCI research aims to find and explain interesting relationships by conducting controlled experiments.
- A cause-and-effect relationship, or a causal relationship, is possible if the experiment is designed and conducted correctly.
- In HCI, the manipulated variable is often a nominal-scale attribute of an interface, and the measured variable is typically a ratio-scale human behavior.
- Finding a causal relationship yields a powerful conclusion that moves the state of the art forward and adds valuable knowledge to the discipline.
- A relationship does not necessarily mean a causal relationship exists, and many relationships are circumstantial.
- Causal relationships emerge from controlled experiments, where participants are selected randomly from a population and are randomly assigned to test conditions.
- Cause and effect conclusions are not possible in certain types of controlled experiments.

4.9 Research topics

- Most HCI research is not focused on designing products or applications.
- HCI research tends to make incremental improvements on existing ideas.
- Big changes in HCI research often involve bringing existing ideas to market through engineering and design.
- Finger flick and two-finger gestures used on touchscreen phones are not new ideas.

- Finding a research topic can be challenging for graduate students in HCI.
- Experienced researchers also struggle with finding the next big idea.
- Four tips will be presented later in the section to overcome "researcher's block".
- Observations will also be presented on ideas and how and where they arise.

4.9.1 Ideas

- The Many Loves of Dobie Gillis was a sitcom about nothing in the post-WWII era
- Each episode began with Dobie on a park bench next to The Thinker statue, where he would come up with his schemes and ideas
- Archimedes is said to have come up with his brilliant idea while taking a bath to determine if King Hiero's crown was pure gold
- Inspiration is not always a single moment of revelation, but can be gradual with unknown sources
- Vannevar Bush's inspiration for memex came from the problem of coping with expanding volumes of information
- Inspiration for ideas in HCI comes from the human experience, as seen in things like "knowledge in the head and knowledge in the world" and metaphor and analogy
- Twitter co-founder Jack Dorsey came up with the idea for the micro-blogging site while sitting on a children's slide in a park eating Mexican food
- Pie menus in graphical user interfaces were first proposed by Don Hopkins at the University of Maryland in 1988

4.9.2 Finding a topic

It is no small feat to find an interesting research topic. In the following paragraphs, four tips are offered on finding a topic suitable for research. As with the earlier dis- cussion on the cost and frequency of errors (see Figure 3-46), there is little science to offer here. The ideas follow from personal experience and from working with students and other researchers in HCI.

4.9.3 Tip #1: Think small!

- A student in HCI is looking for a research topic related to aviation
- His supervisor has funding for a large research project related to aviation
- The broad topic is to develop an improved user interface for an air traffic control system
- The student is unsure where to begin and what the research questions are
- Graduate students are often given big problems to solve by their supervisors' funding sources
- The advice is to narrow down the problem to a small aspect of the interface or interaction
- Pursuing a small idea can lead to related improvements and a dissertation topic
- Don't hesitate to think small.

4.9.4 Tip #2: Replicate!

- Replicating an existing experiment from HCI literature is an effective way to start research.
- Students often get stuck finding a topic for research projects.
- Advising students to study HCI literature and find an experimental research paper on a topic of interest to replicate.
- Replicating a research experiment requires a lot of work and empowers the researcher with a deep understanding of the issues.
- Replicating can lead to a new idea or twist.
- Avoid requiring novelty as it may backfire and bring the process to a standstill.
- Keep expectations low and focus on small tweaks and changes.

4.9.5 Tip #3: Know the literature!

- Reviewing research papers is an excellent way to develop ideas for research projects
- Identify the topic of interest in a general sense and search out and analyze all published research on the topic
- Use resources like Google Scholar, ACM Digital Library, etc. to download and organize papers
- Create a spreadsheet file and tabulate features from the papers in rows and aspects of the interface or interaction technique, conditions tested, results obtained, etc. in columns
- The process is chaotic at first, but the gain is in bringing shape and structure to the chaos
- The table will grow as more papers are found and analyzed
- Similar tables can be found in other research papers
- The point is the process of building such a table, which shapes the research area into relevant categories of inquiry.

4.9.6 Tip #4: Think inside the box!

- "Think outside the box" means to challenge assumptions and beliefs.
- The problem with the challenge is that people often think they already do it.
- The precept of tip #4 is that you are inside the box.
- Thinking inside the box means challenging one's own experiences.
- Observe and question every interaction and situation throughout the day.
- An example is given of pushing the wrong elevator button and analyzing why it happened.
- These experiences can generate ideas for research topics.
- Chapter 4 examines the scientific foundations for research in human-computer interaction.
- The next challenge is designing and conducting experiments using human participants to evaluate new ideas for user interfaces and interaction techniques, explored in Chapter 5.

CHAPTER 5

Designing HCI Experiments

- Designing and conducting experiments with human participants is a necessary skill for HCI researchers
- The signal and noise metaphor is useful for understanding experiment design
- The signal represents the variable of interest, while the noise is everything else that could influence the results
- Noise includes environmental factors and participant variability
- The goal of experiment design is to enhance the signal while reducing the noise
- Proper experiment setup includes careful consideration of manipulated and measured variables, controlled variables, procedures, and tasks
- Experiment methodology is established through these properties.

5.1 What methodology?

- Methodology refers to the way an experiment is designed and carried out
- Having a sound methodology is critical for scientific research
- A standardized methodology strengthens experimental research, while an ad hoc methodology weakens it
- For research in human-computer interaction, the appropriate methodology is plucked from experimental psychology
- The American Psychological Association (APA) is the dominant organization overseeing experimental psychology
- The APA Publication Manual is a valuable resource for researchers undertaking experimental research involving human participants
- The manual lays out many methodological issues such as naming and referring to independent and dependent variables, recruiting participants, reporting the results of statistical tests, and so on
- David Martin's Doing Psychology Experiments is another useful resource for experimental research
- The proceedings of the ACM SIGCHI's annual conference (CHI) are an excellent resource for research papers involving human participants
- Many research papers in the CHI proceedings do not present experimental research
- Of those papers that present a user study, unfortunately, many have an ad hoc methodology
- However, many CHI papers stick to the standard methodology for experiments with human participants
- Examples of such papers from the CHI proceedings are provided.

5.2 Ethics approval

- Ethics approval is a crucial step before the design of every HCI experiment.
- HCI research involves humans, so researchers must respect the safety, welfare, and dignity of human participants.
- The approval process is governed by the institution or funding agency overseeing the research.

- Research projects must be approved by a Human Participant Review Committee (HRPC), Institutional Review Board (IRB), Ethics Review Committee (ERC), or similar committee.
- The review committee ensures that ethical guidelines are acknowledged and adhered to.
- Ethical guidelines include the right of the participant to be informed about the nature of the research, research methodology, risks and benefits, and the right not to participate, not to answer questions, and to terminate participation at any time without prejudice.
- The right to anonymity and confidentiality is also ensured.
- Special attention is given to vulnerable participants such as pregnant women, children, or the elderly.
- The basis for approving research involving human participants is achieving a balance between risks to participants and benefits to society

5.3 Experiment design

- Experiment design is crucial for testing hypotheses in HCI research
- It involves deciding on variables, tasks, procedures, and participant recruitment
- Starting an HCI experiment can be difficult after creative idea generation
- Identifying experimental variables is a useful exercise for transitioning to testable questions
- Independent and dependent variables are the two most important experimental variables
- Independent variables are the factors being compared, such as an existing and a new interface
- Dependent variables are the human behaviors being measured, such as task completion time or steps taken

5.4 Independent variables

- An independent variable is manipulated or systematically controlled to change a human response while the user is interacting with a computer.
- Independent variables are also called factors and are manipulated across multiple levels.
- Independent variables are independent of participant behavior, meaning participants can't influence them.
- Examples of independent variables are device, feedback modality, display size, display type, cross-display technique, transfer function, tree visualization, and navigation technique.
- Independent variables can also be characteristics of humans or environmental circumstances.
- Express independent variables both in terms of the circumstance or characteristic and the levels of the circumstance or characteristic chosen for testing.
- Stick with the chosen terms consistently throughout a paper.
- Experiments often have more than one independent variable.

5.5 Dependent variables

Summary :

- The passage explains dependent variables and their use in Human-Computer Interaction (HCI) experiments. It highlights that the most common dependent variables used in HCI experiments are speed and accuracy, and there are numerous other dependent variables used in HCI experiments. The passage emphasizes that any measurable aspect of human behavior is a potential dependent variable if it can differentiate between two test conditions. The article also provides examples of dependent variables, such as negative facial expressions, read text events, and refocus events. Finally, the passage emphasizes the importance of pilot testing, data collection, and data organization in facilitating follow-up analyses.

Important Points:

- Dependent variables are measured human behaviors that are dependent on the participant's behavior.
- The most common dependent variables in HCI experiments are speed and accuracy.
- Other dependent variables used in HCI experiments include preparation time, action time, throughput, gaze shifts, mouse-to-keyboard hand transitions, presses of backspace, target re-entries, retries, key actions, wobduls, etc.
- Any observable, measurable aspect of human behavior is a potential dependent variable.
- It is important to define, measure, and analyze any human behavior that might expose differences in the interfaces or interaction techniques under investigation.
- Dependent variables should be named separately from their units.
- Pilot testing is crucial to ensure the data collected are correct and available in an appropriate format for follow-on analyses.
- The data should include codes to identify the participants and test conditions.

5.6 Other variables

- There are 3 other variables besides independent and dependent variables
- These are control, random, and confounding variables
- They are often not mentioned in research papers
- Understanding each of these variables is important for experimental research.

5.6.1 Control variables

- There are variables that are not under investigation in experiments.
- Control variables need to be accommodated in experiments.
- Examples of control variables are room lighting, temperature, background noise, display size, mouse shape, mouse cursor speed, keyboard angle, chair height, etc.
- Researchers often don't think about these conditions, but they might influence a dependent variable.

- Controlling the variables means fixing them at a nominal setting during the experiment.
- Extreme values of control variables might interfere with the outcome of the experiment.
- Controlling characteristics of participants, such as right-handedness or 20/20 vision, may be necessary for some experiments.
- Having many control variables reduces variability in measured behaviors but results in less generalizable results.

5.6.2 Random variables

- Random variables introduce more variability in measures but yield more generalizable results.
- Random variables typically pertain to characteristics of the participants such as biometrics, social disposition, or genetics.
- Control variables are circumstances or factors that are fixed at a nominal setting during the experiment to prevent interference with the dependent variable.
- Control variables can pertain to circumstances like room lighting, temperature, background noise, display size, mouse shape, mouse cursor speed, keyboard angle, and chair height.
- Researchers may also need to control participant characteristics in some cases.
- Having lots of control variables reduces variability in measured behaviors but yields less generalizable results.

5.6.3 Confounding variables

- Confounding variables change systematically with an independent variable in experimental research.
- Confounding variables can lead to incorrect interpretation of results.
- Researchers must eliminate, adjust or consider confounding variables to ensure accurate results.
- Camera distance can be a confounding variable in experimental research.
- Confounding variables are sometimes found in Fitts' law experiments.
- Target width is a confounding variable when ID and amplitude are independent variables.
- Practice is a confounding variable in an experiment with interaction technique as an independent variable.
- Prior experience with Google is a confounding variable in comparing two search engine interfaces.
- In text entry experiments, participants must have the same level of experience with both keyboard layouts being compared.

5.7 Task and procedure

-
- Independent variable is a circumstance or characteristic that is manipulated or controlled to elicit a change in a human response while interacting with a computer

- Designing a good task has two objectives: represent and discriminate
- A good task should be representative of actual or expected usage to improve external validity
- A good task should be able to discriminate the test conditions to elicit behavioral responses that expose benefits or problems among the test conditions
- Choosing a task that is too representative of actual usage may compromise the ability of the task to discriminate among the test conditions
- The best task is one that is natural yet focuses on the core aspects of the interaction, the points of differentiation between the test conditions
- Sometimes an independent variable necessitates using a knowledge-based task, which requires changing the task to prevent exposure to the same task
- The experimental procedure includes the task, instructions, demonstration, practice, and questionnaires

5.8 Participants

- Researchers assume that their results apply to people who were not tested
- Two conditions required to apply results to other people: tested people must be from the same population and a sufficient number of participants must be tested
- Variability in measurements affects the likelihood of obtaining statistically significant results, increasing the number of participants increases the likelihood of achieving statistically significant results
- Use the same number of participants as in similar research, using more participants can lead to statistically significant results for a difference of no practical significance
- Not enough participants can prevent statistical significance from appearing
- Power analysis can determine the number of participants required but is rarely done, studying published research is recommended
- Usability evaluations typically don't require a large number of participants, about five participants are sufficient to expose about 80% of the usability problems
- Participants should be selected carefully, and the selection criteria should be stated in the write-up of the methodology
- Participants are usually required to sign a consent form before testing, and their privacy, anonymity, and confidentiality should be protected.

5.9 Questionnaire design

- Questionnaires are a part of most HCI experiments with two purposes: gathering information on demographics and experience with related technology, and soliciting participants' opinions on devices or interaction tasks.
- Questionnaires are the primary instrument for survey research seeking to solicit a large number of people for their opinions and behaviors on a subject.
- Closed-ended questions constrain a participant's response to a small set of options, making follow-on analyses straightforward.
- It is usually important to know the gender and age of participants, since this helps identify the population.
- Age can be solicited as an open-ended ratio-scale response or an ordinal response.
- Questionnaires are also used at the end of an experiment to obtain participants' opinions and feelings about the interfaces or interaction techniques.

- Likert scale is often used to format items in questionnaires to facilitate summarizing and analyzing responses.
- The NASA-TLX assesses perceived workload on six subscales, while the ISO 9241-9 standard for non-keyboard input devices includes a questionnaire with 12 items to assess the comfort and fatigue experienced by participants.
- Items in questionnaires may be tailored according to the device under test.

5.10 Within-subjects and between-subjects

- Test conditions can be either within-subjects or between-subjects.
- Within-subjects, also known as repeated measures, involve each participant being tested on all levels of a factor.
- Between-subjects involve a separate group of participants for each test condition, with each participant only tested on one level of the factor.
- Between-subjects designs require more participants but less testing per participant.
- Within-subjects designs require fewer participants but more testing per participant.
- Practice is an example of a factor that must be within-subjects.
- Gender and handedness are examples of factors that must be between-subjects.
- Within-subjects designs have advantages in terms of variance due to participants' predispositions and not needing to balance groups of participants.
- Between-subjects designs avoid interference between test conditions.
- In HCI experiments, within-subjects designs are preferred due to advantages such as requiring fewer participants and participants' predispositions having a similar variance across test conditions.
- Interference can be an issue in within-subjects designs, but warm-up trials can mitigate this.

5.11 Order effects, counterbalancing, and latin squares

Summary :

- The text discusses the issue of order effects or sequence effects when levels of a factor are assigned within-subjects in an experiment. The interference between the test conditions may result in the improvement of the participants' performance as they progress from one test condition to the next, which is referred to as a practice effect or a learning effect. The text also explains how to compensate for the order effect by counterbalancing, which is a compensatory ordering of test conditions to offset practice effects. The most common method of counterbalancing is to divide participants into groups and administer the conditions in a different order for each group. The text discusses the simplest case of a Latin square, which is an n × n table filled with n different symbols. The balanced Latin square is also explained, which can be constructed for even-order tables. When designing a within-subjects counterbalanced experiment, the number of levels of the factor must divide equally into the number of participants.

Key points:

- Interference between the test conditions may result in order effects
- Practice effect or learning effect occurs when the participants' performance improves as they progress from one test condition to the next
- Counterbalancing is a compensatory ordering of test conditions to offset practice effects
- The simplest case of counterbalancing is a Latin square
- Balanced Latin square can be constructed for even-order tables
- The number of levels of the factor must divide equally into the number of participants when designing a within-subjects counterbalanced experiment.

5.12 Group effects and asymmetric skill transfer

- Within-subjects design: If the learning effect is the same from condition to condition, then the group means on a dependent variable should be approximately equal.
- The point of counterbalancing: the advantage due to practice for a condition tested later in the experiment is offset equally by the disadvantage when the same condition is tested earlier in the experiment.
- A group effect: differences across groups in the mean scores on a dependent variable. It is typically due to asymmetric skill transfer.
- Asymmetric skill transfer: differences in the amount of improvement, depending on the order of testing.
- Example experiment: Compared two types of scanning keyboards for text entry, Letters-only (LO) scanning keyboard with a similar keyboard that added word prediction (L + WP).
- Counterbalancing was used to compensate for learning effects.
- The data showed asymmetric skill transfer.
- The mean for Group 1 was 31.4 cpm, and the mean for Group 2 was lower at 28.8 cpm, an 8 percent performance disadvantage for Group 2.
- Asymmetric skill transfer is usually explainable by considering the test conditions or the experimental procedure.

5.13 Longitudinal studies

- Longitudinal studies involve observing and measuring learning and skill acquisition in participants over a prolonged period.
- In these studies, the amount of practice is an independent variable with each unit of testing serving as a level of the independent variable.
- Performance trends in longitudinal studies often accompany an equation and best-fitting curve demonstrating the power law of learning.
- The goal of a longitudinal study is often to compare the viability of a new technique against current practice, where current practice is any conventional interaction quantifiable by a performance measure.
- New technologies often struggle to overcome the costs of manufacturing, retooling, and user perceptions and attitudes to displace established but less-optimal designs.

- Longitudinal studies have shown that the Dvorak simplified keyboard (DSK) provides a speed advantage over Qwerty, but the cost of retraining users is significant.
- Soft or virtual keyboards provide an opportunity to introduce an optimized design since there is no retooling cost associated with an alternative design.

5.14 Running the experiment

- Pilot testing is useful for smoothing out the protocol and ensuring testing time fits into schedule
- Experiment begins with experimenter introducing and explaining the experiment to participants
- A brief questionnaire is administered to gather demographic data and related experience
- Instructions for tasks should be carefully considered and given to all participants in the same manner
- Experimenter should be neutral and not convey a particular attitude towards the experiment or participants

CHAPTER 6

Hypothesis Testing

- It is advisable to avoid discussions on religion, politics, and statistics at a cocktail party due to strong and divergent opinions on these topics.
- Statistical procedures for hypothesis testing come in two flavors: parametric tests and non-parametric tests.
- Parametric tests operate on data from a probability distribution while non-parametric tests make no assumption about the distribution of the underlying data.
- Parametric tests are applicable to ratio data and interval data while non-parametric tests are applicable to any scale of data.
- The non-parametric procedures examined in this chapter include the chi-square ($\chi 2$) test for nominal data and the Mann-Whitney U, Wilcoxon Signed Rank, Kruskal-Wallis, and Friedman tests for ordinal data.
- ANOVA is the most widely used parametric procedure in experimental research in HCI.
- ANOVA is applicable to ratio data and measures human performance measurements such as time, speed, accuracy, counts for events such as key presses, finger flicks, gaze shifts, or target re-entries.
- Parametric tests are superior to non-parametric tests in terms of power, provided the assumptions are reasonably well adhered to.
- The largest part of this chapter is directed at the ANOVA, and the approach is more cookbook-like than academic.

6.1 Analysis of variance

- ANOVA (F-test) is the main statistical procedure for hypothesis testing in factorial experiments
- Majority of HCI research papers include the results of an ANOVA, giving an F-statistic, degrees of freedom, and associated p-value
- ANOVA determines if independent variable (test conditions) had a significant impact on dependent variable (measured responses)
- ANOVA partitions the variance in the observations on a dependent variable into components attributable to the variance
- ANOVA can be set up differently depending on the experiment design, including simple one-factor design with two test conditions or more complex designs with two or more factors
- Within-subjects assignment of test conditions to participants is assumed in initial examples, but some designs have test conditions assigned between-subjects

6.1.1 Why analyze the variance?

- Research questions are usually comparative to determine the superior performance of two or more interfaces or interaction techniques based on dependent variables such as task completion time, error rate, etc.
- The statistical question pursued is not a question but a statement known as the null hypothesis, which assumes there is no difference in the means.
- ANOVA tests the data to determine the likelihood of the null hypothesis being true or false.
- A statistically significant result means there is little likelihood the null hypothesis is true, and the difference was caused by distinguishable properties in the test conditions.
- A hypothetical experiment is discussed with 10 participants performing a task using two interaction techniques referred to as Method A and Method B, respectively.
- The experiment has a single factor (Interaction Method) with two levels (Method A and Method B).
- Performance is measured on a dependent variable such as task completion time, and the means are computed for each participant over a few trials.
- The difference in the means can be statistically significant or not statistically significant depending on the variability in the observations (variance).
- ANOVA is a statistical tool used to determine the significance of the difference between the means, and its most important statistic is the "P-Value."
- A low probability in the P-value indicates that the differences in the means are unlikely to have occurred in view of the null hypothesis.

6.1.2 More than two test conditions

- ANOVA can involve more than one factor with multiple levels
- An example involves four user interfaces or interaction methods
- ANOVA table shows a significant effect of Test Condition on the dependent variable
- Degrees of freedom are (n-1) for the variance due to Test Condition and (n-1)(m-1) for the variance due to Test Condition × Subject
- P-value in the ANOVA table is .0047, indicating a less than 0.5% chance of chance outcome
- The difference in means was likely caused by inherent properties of one or more of the test conditions
- The effect of the test conditions on the dependent variable was statistically significant

6.1.3 Post hoc comparisons

- ANOVA in Figure 6.10 shows that at least one mean is significantly different from at least one other mean
- Post-hoc comparison test (Scheffé test) used to determine which test conditions differ from each other
- Among the four test conditions, A-C and B-C comparisons are significant while the other four comparisons did not meet the 5 percent threshold for significance
- Several HCI research papers using Scheffé post hoc analysis are mentioned

6.1.4 Between-subjects designs

- Within-subjects designs are more common in HCI research, but between-subjects designs are also used.
- ANOVA analysis must be set up properly for both designs, as the variances are partitioned differently.
- An example experiment is given, where handedness is a between-subjects factor with two levels: left handers and right handers.
- Each participant performed a few iterations of an "add calendar entry" task, and the mean times to complete the task for each participant are tabulated in Figure 6.12a.
- ANOVA is performed on the 16 x 2 matrix in the right two columns in Figure 6.12a, and the result is shown in Figure 6.13.
- The P-value in Figure 6.13 is greater than .05, indicating that the difference between left handers and right handers was not statistically significant.
- Null hypothesis is considered tenable, and it is concluded that there is no performance difference between left and right-handed users in performing the add calendar entry task on the system under test.

6.1.5 Two-way analysis of variance

- An experiment with two independent variables is called a "two-way design"
- Two-way designs analyze the main effects of each factor and the interaction effect
- There are three possibilities for assigning conditions to participants in experiments with two factors: within-subjects for both factors, between-subjects for both factors, or within-subjects for one factor and between-subjects for the other
- The example experiment is a 3 x 2 within-subjects design with three devices and two tasks
- The means and standard deviations for each device-task condition are shown in Figure 6.14a
- Figure 6.14b and c demonstrate the device and task effects in a clearer manner
- An analysis of variance is performed on the core 12 x 6 matrix in Figure 6.14a to determine the significance of the observed differences between devices and tasks
- Figure 6.15 shows the device main effect, task main effect, and device by task interaction effect
- Results should be presented in terms of data rather than statistical methods and should be selectively pulled to improve readability
- Further discussion is necessary to explain the results.

6.1.6 ANOVA tool

- Anova2 is a Java utility available on the book's website
- Anova2 is a command-line application that processes data in a text file and produces an ANOVA table on the console
- Anova2 supports five designs: One-way with one within-subjects factor, One-way with one between-subjects factor, Two-way with two within-subjects factors, Two-way with one within-subjects factor and one between-subjects factor, Three-way with two within-subjects factors and one between-subjects factor

- Figure 6.17 shows the usage message of Anova2 if run without arguments
- The file dix-example-10x2.txt contains data from a hypothetical experiment on icon recognition
- The experiment used 10 participants and measured task completion time as the dependent variable
- The single factor (F1) is icon design with two levels: natural and abstract
- An analysis of variance determines if the difference in task completion time between the two icon designs was statistically significant or likely due to chance
- The output of the analysis is an ANOVA table that reveals a significant effect of icon type on task completion time ($F_{1,9} = 33.36$, $p < .0005$)

6.1.7 Counterbalancing and testing for a group effect

- Learning effects must be offset between conditions if a factor is assigned within-subjects
- Counterbalancing is the most common way to offset learning effects
- For a single factor with two levels (A and B), participants are divided into two equal groups (G1 and G2)
- G1 is tested on A followed by B, while G2 is tested on B followed by A
- The goal is to balance out, or cancel, any learning effect that takes place
- If counterbalancing worked, there should be no group effect—no significant difference between G1 and G2
- A group effect is investigated by treating group as a between-subjects factor
- The experiment is a 2 × 2 design with one within-subjects factor (icon type with levels natural and abstract) and one between-subjects factor (group with levels NA and AN)
- The ANOVA is performed on the data using Anova2
- The group effect is not significant, meaning counterbalancing worked
- The result is still a significant effect of icon type on task completion time
- There is a two-way interaction effect between icon type and group, which is not significant
- In many research experiments, counterbalancing is used for within-subjects factors without actually testing for a group effect.

6.2 Chi-square test

- Chi-square test, also known as the Pearson chi-square test, is a statistical procedure to investigate relationships between categorical or nominal-scale variables representing attributes of people, interaction methods, systems, etc.
- Data is commonly summarized in a contingency table, which is a cross-tabulation organizing data in rows and columns with each cell containing counts or frequency data for the number of observations in the category.
- A chi-square test compares the observed values (counts in the table) against expected values developed under the assumption that there is no difference among the categories in the table.
- The chi-square test is a non-parametric test since the categories are nominal-scale attributes and do not have a probability distribution associated with them.

- A hypothetical example is given to illustrate how the chi-square test is used to determine if the observed differences between males and females in their methods of scrolling on desktop computer systems are statistically significant.
- The test statistic is written $\chi 2$, using the lowercase Greek letter chi, and it is a comparison between the observed values and the expected values, with the expected values accounting for the different row and column totals.
- Each chi-square is the square of (the observed value minus the expected value), divided by the expected value.
- The final chi-square value is the sum of the individual chi-squares in the table.
- Determining if there is a statistically significant difference among the cell entries involves comparing the final chi-square with values in a table of critical values.
- The degrees of freedom for the chi-square statistic is (r-1)(c-1), where r is the number of rows and c is the number of columns.
- The required alpha level, or p-value, for significance is chosen prior to testing.
- The critical value for significance is looked up in a table.
- The book's website includes a ChiSquare Java utility that automates the calculation of the chi-square statistic and calculates the probability p.

6.3 Non-parametric tests for ordinal data

- Non-parametric tests make no assumptions about probability distribution of data
- Non-parametric tests are applicable to a wider range of data than parametric tests
- Non-parametric tests deal with ordinal data (ranks) while parametric tests operate on interval or ratio data
- Non-parametric tests ignore any property of the scale of data except ordinality
- Non-parametric tests sacrifice the power to use all available information to reject a false null hypothesis
- Non-parametric tests are typically used in HCI research to analyze ordinal (and sometimes interval) data obtained subjectively, from participants, or using heuristics or other non-empirical or semi-empirical methods
- Four most common non-parametric procedures are the Mann-Whitney U test, the Wilcoxon Signed-Rank test, the Kruskal-Wallis test, and the Friedman test, each used in a particular context depending on the number of test conditions and experiment design
- Between-subjects designs generate independent samples while within-subjects designs generate correlated samples, as shown in Figure 6.29
- Kruskal-Wallis and Friedman tests operate on three or more conditions and are usually followed with a post hoc pairwise comparisons test if statistically significant
- Examples of non-parametric tests in action with data analysis provided using StatView and a Java utility on the book's website.

6.3.1 Example 1

- Researcher wants to determine political leanings of Mac and PC users
- 10 Mac users and 10 PC users are randomly selected and assessed on a 10-point scale
- Data is potentially interval-scale, but researcher decides to use non-parametric test
- Mann-Whitney U test is chosen due to two conditions and between-subjects assignment
- Results show $U = 31.0$ with $p = .1418$ (corrected for ties)
- Conclusion is that there is no statistically significant difference in political leanings between Mac and PC users ($p > .05$).

6.3.2 Example 2

- A researcher compares two designs of media players for their "cool appeal" among young users.
- Ten tech-savvy participants rate the designs on a 10-point linear scale.
- Media player A has a mean rating of 6.4, while media player B has a mean rating of 3.7.
- A statistical test is used to determine if the differences in ratings are significant.
- Since the data is interval-scale, a non-parametric test (Wilcoxon Signed-Rank) is chosen.
- The test shows a normalized z score of -2.254 and a p-value of 0.0242 (corrected for ties).
- Since $p < 0.05$, the two media players are found to differ significantly in "cool appeal."

6.3.3 Example 3

- The designers of a new GPS system for automobiles want to know if age will affect its acceptance.
- They recruit 8 participants from three age groups (20-29, 30-39, and 40-49).
- Participants rate the new GPS system on a 10-point scale and indicate if they would consider purchasing it.
- Mean scores by age group are 7.1 (20-29), 4.0 (30-39), and 2.9 (40-49).
- There appears to be a difference in scores among the three age groups.
- A non-parametric Kruskal-Wallis test is used to determine if the differences are statistically significant.
- The test statistic is H, with a chi-square distribution of $df = k-1$ (k is the number of groups).
- The test shows a significant effect of age group on the acceptability of the GPS design ($\chi 2 = 9.605, p<.01, df=2$).
- A post hoc pairwise comparisons test is needed to determine which age groups are significantly different from one another.

6.3.4 Example 4

- The researcher has implemented four variations of a search engine interface (A, B, C, and D)
- The researcher wishes to know if the interfaces differ in the quality of results they produce
- Eight participants are recruited and are given a demonstration of each interface and then asked to do a series of search tasks
- The “quality of results” is assessed for each participant and for each search engine on a linear scale from 1 to 100 (1 = very poor, 100 = very good)
- The mean scores for the search engines from A to D were 71.0, 68.1, 60.9, and 69.8
- The Friedman test is chosen since the data is interval-scale and there are four test conditions assigned within-subjects
- The Friedman test shows that there is a difference in the quality of results between the four search engine interfaces
- A post hoc pairwise comparisons test is used to determine which search engine interfaces are significantly different from one another
- Two pairwise comparisons were found to be significant: 1:3 and 3:4, i.e., between interfaces C and A, and between interfaces C and D.

6.3.5 Discussion

- Mean is the correct statistic for central tendency when the data is interval-scale
- Response items on questionnaires may not be interval-scale if they are subject to human interpretation
- Non-linear scales compromise the quality of data and degrade to ordinal, making median or mode the appropriate statistic for central tendency
- Different researchers hold different opinions on which measure is the correct measure
- Non-parametric tests are limited to single-factor analyses and cannot be used for multi-factor analyses
- Non-parametric tests are commonly used to analyze questionnaire data or ratings
- Non-parametric tests are fine for gathering participants’ comments and preference ratings on the conditions tested
- Non-parametric tests for ratio-scale data will be discussed in the next section.

6.4 Parametric versus non-parametric tests

- When choosing among alternative tests to study data in an experiment, a rationale is needed.
- The guiding principle for statistical tests is power, which is about improving the odds of getting it right with the null hypothesis.
- An appropriate test is one where the measurement requirements of the test are consistent with the measurements used in the research.
- Parametric tests assume the data are sampled from a probability distribution, such as the normal distribution.

- If the assumptions for parametric tests are not met, the researcher must decide on the appropriate course of action.
- The three possibilities for when assumptions for parametric tests are not met are: (1) proceed with the parametric test, (2) transform or clean the data in some manner to correct the violations and then proceed with the parametric test, or (3) use a non-parametric test.
- For ordinal data, non-parametric tests are the logical choice.
- For ratio data, parametric tests are preferred, in part due to their increased power.
- Non-parametric tests collapse data into ranks, sacrificing information.
- Choosing a non-parametric test for ratio-scale measurements that deviate from normality is to replace one deficiency with another.

CHAPTER 7

Modelling Interaction

- A model is a simplification of reality
- Models allow exploration of phenomena, changes and testing without actually constructing or throwing the object
- This chapter is about modeling interaction, building, testing and using models, and thinking about interaction through models
- The term model is used loosely and varies depending on the field
- There is a continuum for models: descriptive models using “loose verbal analogy and metaphor” describe phenomena, while predictive models use “closed-form mathematical equations” to predict phenomena
- Descriptive models provide insight into a design or interaction problem, while predictive models are used to make predictions
- Examples of descriptive and predictive models are presented in the chapter.

7.1 Descriptive models

- Descriptive models are common in HCI research.
- They are often developed naturally without being explicitly called models.
- Examples include design spaces, frameworks, taxonomies, and classifications.
- These models provide insight into design and interaction problems.

7.1.1 Delineating a problem space

- A descriptive model simplifies a problem space by dividing it up into constituent parts or processes
- Without delineation, the problem space is vast and uncharted
- Descriptive models can be found in HCI papers with titles or section headings such as design space, framework, taxonomy, or classification
- A non-HCI example of a descriptive model is Johnson's chart for politics
- Johnson's chart is organized as a semi-circular spoked wheel with politics at the center
- Descriptive models add tremendous insight into a problem and empower critical thinking
- In HCI, descriptive models are used to analyze problem spaces and inform interaction design

7.1.2 Quadrant model of groupware

- Computer supported collaborative work (CSCW) is an important area of research in HCI
- Applications that support collaboration are called groupware
- Groupware can be divided into simpler components or processes through differences in aspects of collaboration
- The quadrant model of groupware is a descriptive model that divides collaboration into spatial and temporal aspects

- The quadrant model presents groupware as a 2x2 matrix with four cells, each representing a distinct quality of interaction
- The quadrant model can be used to identify common elements for interactions located in the same area of the problem space
- The quadrant model was published in 1991 and may not reflect current forms of collaboration, but it provides a context for analysis.

7.1.3 Key-action model (KAM)

- Computer keyboards contain a vast array of keys
- Keyboard keys are categorized into symbol keys, executive keys, and modifier keys
- Symbol keys deliver graphic symbols to an application (letters, numbers, or punctuation symbols)
- Executive keys invoke actions in the application or at the system level (e.g., enter, F1, and esc)
- Modifier keys set up a condition that modifies the effect of a subsequently pressed key (e.g., shift and alt)
- The KAM model identifies three categories of keys and provides a name, a definition, and examples for each category
- Figure 7.3 illustrates a keyboard with keys highlighted according to the KAM
- The KAM model can be used to expose problems and suggest opportunities in keyboard design
- The analysis using KAM shows a right-side bias in desktop keyboards, with a 3:18 left-right ratio of executive keys
- The right hand is overloaded with the mouse and executive-key activation, posing a problem for right-handed users
- Left-handed mouse usage is more efficient with right-side power-key operations
- Key-ambiguity descriptive model might be useful in analyzing keyboards where a key can produce two or more symbols
- Descriptive models like KAM and key-ambiguity model can be used to tease out design issues and improve keyboard design.

7.1.4 Model of bimanual control

- Humans use each of their hands differently, and this has undergone considerable study in an area of human motor control known as bimanual control or laterality.
- Most tasks are asymmetric in nature.
- Guiard proposed a three-part descriptive model of bimanual control, which identifies the roles and actions of non-preferred and preferred hands.
- The model has been widely adopted in human-computer interaction (HCI) research in two-handed interaction.
- Scrolling is traditionally accomplished by dragging the elevator of the scrollbar positioned along the right-hand side of an application's window, which is a target acquisition task.
- The introduction of the scrolling wheel between mouse buttons in 1996 put scrolling in the preferred hand, increasing right-side bias.

- Guiard's model presents an opportunity for design by analyzing scrolling and accompanying tasks and revealing that scrolling is an appropriate task for the non-dominant hand.
- An analysis of scrolling and the accompanying tasks reveals that scrolling is well suited to the non-preferred hand.
- Scrolling is an appropriate task for the non-dominant hand, and this is revealed through Guiard's model of bimanual control.

7.1.5 Three-state model for graphical input

- Buxton’s three-state model of graphical input is a descriptive model for pointing devices
- The model identifies three states: Out of Range, Tracking, and Dragging
- The model can be extended to capture additional aspects of pointing device interaction
- MacKenzie and Oniszczak used the three-state model to help redesign an interaction technique used in the Blackberry Storm and Apple MacBook
- Touchpads are the dominant pointing device for notebook computers and support "lift-and-tap" interaction technique
- Buxton's model can represent the state transitions for dragging tasks using a mouse and using lift-and-tap on a touchpad
- The use of lift-and-tap necessitates extra state transitions and can be error-prone
- The state transitions on touchpads were redesigned using pressure sensing capability and feedback to inform the user of state transitions
- Buxton’s model continues to serve as a descriptive model for interactive systems, with contemporary applications including models for preview and undo, puck and stylus interactions, camera control, modeling multi-touch on touchscreens, modeling panning and zooming on touchscreens, modeling the selection of moving targets, and describing the rotation mode of a three DOF mouse.
- Predictive models work with closed-form mathematical equations.

7.2 Predictive models

- A predictive model is an equation that predicts the outcome of a variable based on the value of one or more other variables (predictors).
- The outcome variable is typically the time or speed in doing a task, or any other measure of human behavior that uses continuous or ratio-scale data.
- The predictor variable is usually referred to as the independent variable, but this terminology is problematic in experimental research, where independent variables have a different meaning.
- Independent variables in experimental research are typically nominal-scale attributes and cannot serve as predictors in prediction equations.
- However, independent variables in experimental research can also be ratio-scale attributes, and such variables can serve as predictors in prediction equations.
- Predictive models are useful in HCI because they allow a problem space to be explored using numbers, not concepts.

- Card, English, and Burr likely presented the first predictive model in HCI in 1978, in which they built and compared Fitts' law models for the mouse and joystick.
- Their work went beyond a typical user study by embedding a theoretical account of the underlying human processes in their models and using them as prediction tools for follow-up analyses of design alternatives.
- Building a prediction equation involves selecting appropriate predictor variables, collecting data on the outcome variable and the predictor variables, and using statistical methods to fit the data to the equation.

7.2.1 Linear regression model

- The basic prediction equation expresses a linear relationship between an independent variable and a dependent variable.
- The equation is $y = mx + b$, where m is the slope and b is the y intercept.
- The best-fitting straight line for the equation is obtained through the process of linear regression.
- To build the equation, a set of x-y sample points is needed.
- The sample points scatter around the line, and the quality of the model depends on how close the points are to a straight line.
- The relationship assumed is linear, which may not always be the case.
- A simpler question is usually asked before building a prediction equation: is there a relationship between the two variables?
- A scatter plot can help visualize the data and determine whether a relationship exists.
- A coefficient of correlation is calculated to measure the strength of the relationship.
- A prediction equation is built to give the best estimate of y in terms of x.
- The amount of variation in the data that is explained by the model is interpreted as R2.
- The standard error of estimate (SE) establishes confidence intervals around a prediction.

7.2.2 Fitts' law

- Fitts' law is one of the most widely used models in HCI.
- It is used for interaction that involves rapid-aimed movements such as moving a finger or cursor to a target and selecting the target.
- Fitts' law has three uses in HCI:
    1. To learn if a device or interaction technique conforms to the model by building the prediction equation and examining the correlation for "goodness of fit"
    2. To use the prediction equation in analyzing design alternatives
    3. To use Fitts' index of performance (now throughput) as a dependent variable in a comparative evaluation.
- Fitts' law comes to HCI by way of basic research in experimental psychology.

7.2.2.1 Background

- Fitts was an experimental psychologist interested in applying information theory to human behavior in the 1950s.
- Fitts proposed that the amplitude of an aimed movement was analogous to the information in an electronic signal and that the spatial accuracy of the move was analogous to electronic noise.
- Fitts proposed that the human motor system is like a communications channel, where movements are like the transmission of signals.
- Fitts' analogy is based on Shannon's Theorem 17.
- Fitts presented his analogy, now known as "Fitts' law," in two highly celebrated papers in 1954 and 1964.
- Fitts examined the relationship between the amplitude of the movement task, the time taken, and the width of the region within which the move terminates.
- Fitts proposed to quantify a movement task's difficulty using information theory by the metric "bits."
- Fitts proposed to quantify the human rate of information processing in aimed movements using bits per second as the units.
- Fitts' index of performance, now called throughput, is calculated by dividing ID (bits) by the mean movement time, MT (seconds), computed over a block of trials.
- If using Fitts' law as a predictive model is the goal, then the movement time (MT) to complete a task is predicted using a simple linear equation.

7.2.2.2 Example (skip)

7.2.3 Choice reaction time

- Hick-Hyman law and Fitts' law arrived in HCI from experimental psychology research on modeling human behavior according to information processing principles
- Hick-Hyman law takes the form of a prediction equation for choice reaction time, given n stimuli associated with n responses
- The time to react (RT) to the onset of a stimulus and choose the correct response is given by $RT = a + b \log_2(n-1)$
- The constants a and b are empirically determined and reasonable values are $a \approx 200$ ms and $b \approx 150$ ms/bit
- The log term in the equation is analogous to the information content of the task and has units of "bits"
- If some of the choices are more probable than others, the information content of the task is reduced, and so is the choice reaction time
- The information (H) for a set of alternatives with different probabilities is given by $H = -\Sigma p_i \log_2(p_i-1)$, where p_i is the probability of occurrence of the ith item in the set
- The Hick-Hyman law has surfaced in a few contexts in HCI research, such as telephone operator selection, measuring and predicting the time to select items in hierarchical menus, and modeling the perception, planning, and activation time for users to switch modes with their non-dominant hands in a tablet interface

- The law has failed to gain the same momentum in HCI as Fitts' law because additional behaviors coincident with choice reaction, such as movement or visual search, complicate applying the model.

7.2.4 The keystroke-level model (page 277)

- The Keystroke-Level Model (KLM) is a predictive model used to analyze human performance in interactive computing systems.
- It was developed as a practical design tool for predicting expert error-free task completion times.
- The model requires the following elements to predict task completion time: Task or series of sub-tasks, Method used, Command language of the system, Motor skill parameters of the user, and Response time parameters of the system.
- The KLM is useful in situations where the sequence of interactions in performing a task is known.
- The model works with four motor-control operators (K, P, H, D), one mental operator (M), and one system response operator (R).
- The total predicted time is the sum of the subtask times.
- Each operation is assigned a value according to the skill of the user.
- The KLM is not useful if the details of the interactions are not known.
- A related model, GOMS, operates at a higher level and attempts to predict the method the user will adopt.

7.2.4.1 Original KLM experiment

- The Keystroke-Level Model (KLM) is one of the earliest and most comprehensive predictive models in the HCI literature.
- The KLM predicts expert error-free task completion times using task, method used, command language of the system, motor skill parameters of the user, and response time parameters of the system.
- The KLM can predict the time for different methods of doing a task.
- The KLM requires a task to be broken down into a series of subtasks or primitive operations.
- The KLM works with four motor-control operators (K = keystroking, P = pointing, H = homing, D = drawing), one mental operator (M), and one system response operator (R).
- To validate the KLM, Card, Moran, and Newell conducted an experiment using 14 tasks performed using various methods.
- The required sequence of subtasks for one of the tasks was given, along with the operators for each subtask.
- The task required four mental operations (M) and 15 keystroking operations (K).
- Card, Moran, and Newell conducted an experiment to validate the KLM using 14 tasks performed with different methods
- The experiment was performed on the POET system for task T1, which required 4 mental operations and 15 keystroking operations
- tM was set to 1.35s and tK was set to 0.23s, and the predicted execution time for task T1 on POET was 8.85s

- Twelve users performed the task an average of 27 times each, and the mean execution time was 7.8s with a deviation of approximately 11% from the prediction
- The experiment validated the KLM, and Figure 7.21 shows the observed versus predicted task execution times for all 14 tasks
- The KLM is a predictive model that can be molded and used for a parametric analysis or sensitivity analysis to explore how predictions are affected by changes in the parameters
- Figure 7.22 is an example of a sensitivity analysis for an editing task that charts texecute for three interaction methods as a function of n (the distance in words from an initial location to the location of a misspelled word to correct)

7.2.4.2 Modern applications of the KLM

- Early experiments with the KLM were focused on text editing and document management tasks.
- Tasks in early experiments included deleting a line of text, adding a label to a box, and transferring a file to another computer.
- Today, tasks like transferring a file to another computer involve a mix of mouse and keyboard operations with a GUI-based FTP application.
- The KLM is more than 25 years old and some of the operators need updating.
- The pointing operator (P) has a constant of 1.10 seconds.

a constant of 1.10 seconds. With the ubiquity of the mouse in today's GUIs, this operator could benefit from the additional fidelity afforded with a Fitts' law prediction. Using the mouse model from Figure 7.17, for example, a reasonable substitute for P is

$$t_P = 0.159 + 0.204 \times \log_2\left(\frac{A}{W} + 1\right) \tag{19}$$

If there are numerous mouse operations in a task undergoing KLM predictions, then the added precision in Equation 19 will improve the analysis. For example, if a subtask involves clicking a 1.2 cm wide toolbar button, having just clicked a neighboring button 3.2 cm away, then the pointing time is[18]

$$t_P = 0.159 + 0.204 \times \log_2\left(\frac{3.2}{1.2} + 1\right) = 0.45\ \text{s} \tag{20}$$

Alternatively, if the mouse pointer must move, say, 44.6 cm to reach the same toolbar button, the pointing time is

$$t_P = 0.159 + 0.204 \times \log_2\left(\frac{44.6}{1.2} + 1\right) = 1.22\ \text{s} \tag{21}$$

The predictions above include the terminating mouse-button click to select the target, unlike the original P operator.

As an example of applying the KLM to current GUI interaction, consider the editing operations to change the font style and font family for text while word processing. Figure 7.23 shows the operations to change "M K" to boldface in the Arial font.

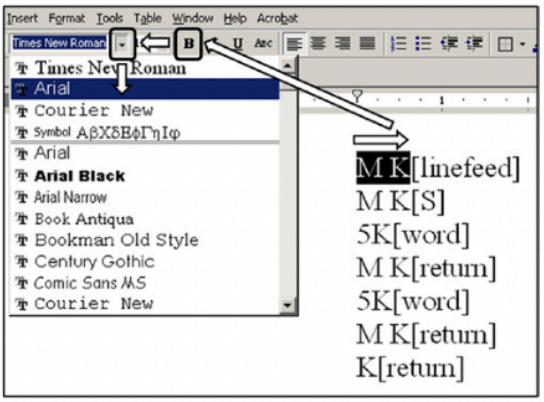

FIGURE 7.23

Mouse operations to change "M K" to boldface in the Arial font. The application is MS Word.

[18]For Fitts' law studies with the mouse, pointing times of 500 ms or less are common for easy tasks ($ID < 3$ bits). The minimum mouse pointing time in the original KLM was 800 ms (see Figure 7.20) which seems high.

The image is a screen snap for the editing performed above using Microsoft Word to characterize task T1. Four pointing operations are required: select the text, select Bold, select the drop-down arrow in the Font list, and select Arial.

A KLM coding of the subtasks is given in Figure 7.24.

Here, the pointing operator appears as P[A,W] to specify the movement amplitude and target width of the pointing task. The values shown were obtained using a ruler applied to Figure 7.23.[19] The execution time for each P operation is given in the right-hand column, computed using Equation 19. As there are four distinct subtasks, each is coded with a mental operator (M) as well as a pointing operator. The predicted task execution time is

$$t_{\text{EXECUTE}} = 4 \times t_{\text{M}} + \sum t_{\text{P}} = 4 \times 1.35 + 2.71 = 8.11\,\text{s} \qquad (22)$$

As with the keystroking operator (K) in the original KLM, applications of the model using P[A, W] should use a pre-test to determine the coefficients in the Fitts' law model for the pointing device and method involved.

Of course, the same task can be done using the keyboard. A set of keyboard subtasks achieving the same effect is shown in Figure 7.25.

Here we see the same four mental operators; however, there is an assortment of keystroking actions, as per Microsoft Word. The KLM uses a separate keystroking

Mouse Subtasks	KLM Operators	t_P (s)
Drag across text to select "M K"	**M P**[2.5, 0.5]	0.686
Move pointer to Bold button and click	**M P**[13, 1]	0.936
Move pointer to Font drop-down button and click	**M P**[3.3, 1]	0.588
Move pointer down list to Arial and click	**M P**[2.2, 1]	0.501
	$\sum t_P$ =	2.71

FIGURE 7.24

Mouse subtasks and KLM operators for editing operations in Figure 7.23.

Keyboard Subtasks	KLM Operators
Select text	**M P**[shift] 3**K**[→]
Convert to boldface	**M K**[ctrl] **K**[b]
Activate Format menu and enter Font sub-menu	**M K**[alt] **K**[o] **K**[f]
Type *a* ("Arial" appears at top of list)	**M K**[a]
Select "Arial"	**K**[↓] **K**(Enter)

FIGURE 7.25

Keyboard subtasks and KLM operators for the task in Figure 7.23.

[19]If the measurements are done on a display, they vary according to the display size and the zoom-factor of the document view. Since target amplitude and width appear as a ratio in Fitts' index of difficulty, scaling up or down does not affect the Fitts' law predictions.

value for pressing a modifier key, such as SHIFT. The predicted execution time using the keyboard is

$$t_{\text{EXECUTE}} = 4 \times t_{\text{M}} + 12 \times t_{\text{K}} = 4 \times 1.35 + 12 \times 0.75 = 14.40\,\text{s} \qquad (23)$$

- Figure 7.26 shows the sensitivity of the prediction to changes in the keystroking time, tK.
- The keyboard method is slower than the mouse method over all but the fastest three settings for tK.
- A separate Fitts' law dragging model is required because dragging operations are less efficient than pointing operations.
- Many contemporary applications use the KLM to model interactions in novel HCI contexts.
- KLM is a malleable tool. It is used, modified to suit the problem.
- Validation is required in a formal or semi-formal experiment where the predicted and observed times are compared.
- Predictive models like the KLM are tools for thinking, for generating insight.
- Reacting, planning, pausing, and attending are hidden costs contributing to deviations between predictions and observations.
- The mental operator (M) in the KLM is its Achilles heel because it is a single operator pegged at 1.35 s and is used for all mental processes.

7.2.4.3 The KLM and (Predictive) text entry

- Keystroke-level modeling (KLM) is a model for expert users.
- KSPC (keystrokes per character) is defined as the number of keystrokes required for each character of text entered using a given input method in a given language.
- The relevant statistic for text entry is KSPC, and it differs for different input methods.
- The KLM prediction does not account for the performance costs of visually attending to the interface.
- For mobile phones, text entry using multi-tap requires more keystrokes than using predictive text entry (T9).
- Interaction with a mobile phone keypad is different from two-handed typing on a computer keyboard.
- For predictive text entry, systems that involve word completion, word prediction, phrase prediction, or other language-based adaptive or predictive features are known to tax the attention of users.
- Predictive text entry on a mobile phone (T9) requires one keystroke per character, and the system uses an internal dictionary to map candidate words to the key sequence.
- Predictive text entry requires the KLM’s mental operator (M) or something similar to model it.

7.2.4.4 Updating the KLM’s mental operator (M)

- The KLM’s mental operator (M) is a single operator and the theory supporting it is unclear.
- Card et al.’s model human processor suggests two additional mental operators - MP (physical matching) and MV (visual searching) that may require attention and cognition.
- The value of M is estimated by removing the predicted time for all physical operations from the observed execution time.

- MP and MV have different execution times and do not have a nominal value like other operators.
- Nominal execution times for the top four mental operators range from 240-566 ms.
- The visual search operator (MV) execution time depends on the number of choices the user must scan.
- The addition of new operators that are more aligned with the user's perceptual and cognitive processes should help with KLM modeling of interfaces that include adaptive or predictive features.

7.2.5 Skill acquisition

- Relationship between skill and practice is well-established
- Novices start with poor performance but improve with practice
- Power function equation (y = bx^a) is a good model to predict the relationship between skill and practice
- The relationship between skill and practice is called the power law of learning or the power law of practice
- The power law of learning dates back to at least the 1920s
- Trial is any practice indicator, such as task number, hours of practice, session number, etc.
- Dependent variable can be time to do a task (T) or speed (S)
- If T is the dependent variable, the equation can be recast as T = Tn^a
- If S is the dependent variable, the equation can be recast as Sn = S1 x n^a
- Opti layout was compared to Qwerty layout for text entry using a stylus
- Crossover point occurred at session 11 where Opti became faster than Qwerty
- Power function equation was used to model the data for both layouts
- Squared correlation coefficient (R2) for Opti was 99.7% and 98.0% for Qwerty, indicating excellent predictability

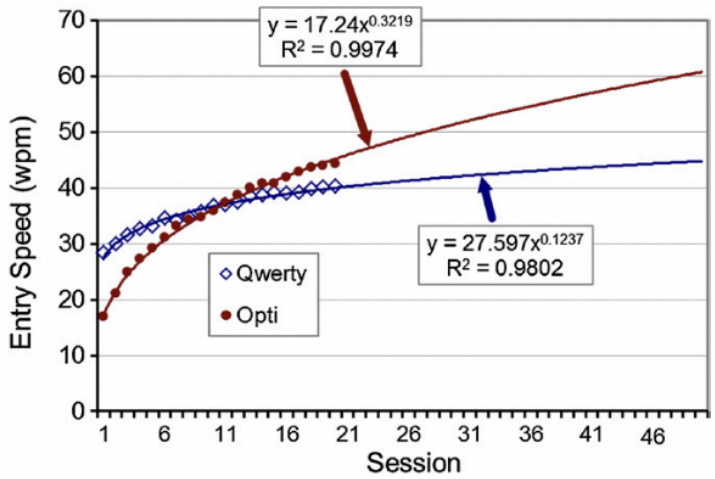

FIGURE 7.37

Power law of learning for Opti and Qwerty showing extrapolation to session 50.

Figure 7.37 also shows an extrapolation of the models to session 50. However, extrapolating beyond the testing range is risky. Although the correlations are high for both models, the ability to predict text entry speed diminishes the farther one ventures from the range of testing. For example, at session 50 the predicted text entry speed for Opti is:

$$S_{50} = 17.24 \times 50^{0.3219} = 60.7 \text{ wpm} \tag{29}$$

That's fast! Too fast, perhaps. Here's a simple exercise to demonstrate a model's ability (or inability!) to extrapolate. Using only the data for sessions 1–10 for Opti, the prediction equation is:

$$S_n = 17.03 \times n^{0.3319} \tag{30}$$

Using this equation to predict the text entry speed at session 20 yields:

$$S_{20} = 17.03 \times 20^{0.3319} = 46.03 \text{ wpm} \tag{31}$$

But the observed text entry speed at session 20 was 44.29 wpm (see Figure 7.36a). The prediction is generous by (46.03 – 44.29) / 44.28 = 4.0%. So perhaps the prediction for session 50 is generous as well.[30]

[30]Perhaps the prediction for Opti at session 50 for 60.7 wpm is realistic. In the experiment, text phrases were drawn at random from a small set of just 70 phrases (≈25 characters/phrase). The number of phrases per session was 50–60 early on, and 90–110 in later sessions. In view of this, participants were likely developing muscle memory with the phrases as the experiment progressed. This is similar to the highly over-learned motor skill one develops in other activities, such as buttoning a shirt, tying shoelaces, or entering a password (for people who never change their passwords). While this argument may have merit, it also means that the predicted text entry speed for session 50, while accurate, may apply only to the phrase set, not to English.

7.2.5.1 Skill acquisition research in HCI

-
- Power law of learning in HCI literature
- Card, English, and Burr (1978) conducted experiment comparing mouse, joystick, and two keyboard techniques for selecting text on CRT display
- Log transformation performed on both x-axis data (block) and y-axis data (positioning time)
- Relationship becomes linear after log transformation
- Relationship for mouse shown in Figure 7.38
- Recker and Pitkow (1996) report similar log-log models for mapping document access to document storage in multi-media database
- Seibel reports log-log model for a choice reaction task (reported in Newell 1990, 7)
- Skill acquisition models for reaction time (pp. 57-59) and execution time (pp. 284-285) in The Psychology of Human-Computer Interaction (Card et al., 1983)
- Numerous other sources in HCI literature reporting predictive models of skill acquisition

7.2.6 More than one predictor

- Equations can predict outcomes of a dependent variable from a single predictor (independent variable).
- The technique of using multiple predictors is called multiple regression.
- A hypothetical example with the relationship between time taken to learn a computer game and the age and computing habits of players is used to illustrate multiple regression.
- The experiment had 14 participants ranging in age from 16 to 40 years, with daily computer use from two to eight hours per day.
- Two scatter plots show a strong relationship between age and time taken to learn the game and between daily computer use and time taken to learn the game.
- A prediction equation using multiple regression can be created with a statistics application.
- The prediction equation with multiple predictors for the example data is $y = 3.2739 + 0.1575x1 - 0.4952x2$.
- MacKenzie and Ware (1993) studied the effect of lag on human performance while using a mouse in point-select tasks.
- The dependent variable was the movement time (MT in ms) to complete point-select tasks.
- Four models were presented for MT. The fourth model demonstrated an observed interaction effect between lag and task difficulty.
- The fourth model reduced to MT = 230 + 169 x IDe at lag = 0.

7.3 A model continuum model

- The problem space of modeling is treated as a dichotomy between descriptive models and predictive models.
- This dichotomy is in conflict with the model continuum suggested by Pew and Baron.
- The author proposes another model, a Model Continuum Model (MCM).
- The MCM shows a continuum with identifiable points along the way.
- Figure 7.42 is a work in progress and lacks organization and clarity.
- The MCM places descriptive and predictive models at opposite ends of the continuum.
- The MCM suggests positions along the continuum where certain activities or processes might be placed.
- Qualitative and quantitative terms may be added at each end of the continuum.
- The chapter presented core ideas of descriptive and predictive models of human-computer interaction.
- The student exercises that follow extend the ideas presented in the chapter.

CHAPTER 8

Writing and Publishing a Research Paper

- Publication is the final step in research
- Final chapter is about writing and publishing a research paper
- Most HCI research papers present a novel interface idea and describe an evaluation in a user study
- User studies should be designed according to accepted standards for experiments with human participants
- Adhering to accepted standards is important in HCI and brings consistency to the field
- Consistency in practice extends to the final step in research - preparing and publishing the results.

8.1 Conference papers, journal papers

- Main venues for publishing HCI research papers are conference proceedings or journals
- Journals are considered top tier for research publications due to tightly controlled peer review and archiving in major libraries
- HCI conferences have emerged as an important vehicle for presenting and publishing research due to short timeline and archival advantage of online databases
- Manuscript is the term for research paper before publication and venue for submitting is decided before writing begins
- Journals have relaxed requirements for format since revisions are introduced before publication
- Conferences have strict requirements on format, including page length, due to short timeline and author responsibility for final camera-ready copy
- Major parts of a research paper for conference or journal submissions describing a user study include a novel interface idea and evaluation according to accepted standards for experiments with human participants

8.2 Parts of a research paper

- The text will examine the constituent parts of a research paper
- Figure 8.1 is used as a backdrop to show the major sections of a 10-page conference paper by Tinwala and MacKenzie (2010)
- The sections in Figure 8.1 apply to the majority of research papers in HCI, but not all
- Some HCI research papers do not describe a user study
- The annual ACM SIGCHI conference welcomes other types of submissions
- The important role of non-experimental research in HCI is worth noting
- The discussion here focuses on a traditional HCI research paper with a user study

8.2.1 Title

- A title must be concise and every word must contribute to it
- The title should identify the subject area and narrow the scope of the work
- Adding a secondary title can either broaden or narrow the scope of the work
- An overly broad title can mislead the readers and fail to indicate the sub-topic of the research
- The author's names, affiliations, and contact information follow the title and are provided according to the submission requirements of the conference or journal.

8.2.2 Abstract

- The abstract is written last and is typically limited to 150 words.
- The abstract should be concise and should not include any citations.
- The abstract is the first part of the paper that is read after the title.
- The objective of the abstract is to tell the reader what was done and what was found.
- A well-crafted abstract should have perfect English and grammar, and the content should be clear and succinct.
- The abstract is not an introduction to the subject matter of the paper.
- The abstract should give specific results and convey the most salient finding(s).

8.2.3 Keywords

- Keywords are used for database indexing and searching
- Keywords are chosen by the author and identify the subject matter and scope of the work
- Research papers published in ACM conference proceedings or journals are required to include indexing and retrieval information according to the ACM's Computing Classification System (CCS)
- CCS includes "Categories and Subject Descriptors" and "General Terms" provided by the ACM, not the author
- Choosing descriptors and terms can be challenging, but mimicking those used in a similar paper in the same proceedings or journal is recommended by the ACM
- The formatting of descriptors and terms must be strictly followed, and care should be taken in the event of an inappropriate descriptor in a published paper

8.2.4 Introduction

- The opening section of a research paper is typically called Introduction or Background.
- The introduction gives the context for the research, characterizes the state of the art, and indicates why the subject matter is interesting and relevant.
- An overview of the contents of the entire paper is usually given within the first page or so of the introduction.
- It is desirable to state the contribution of the work without using a strong or boastful statement.

- The introduction may span several sections with any reasonable organization.
- A literature review is typically organized in a separate section with an appropriate heading.
- The literature review discusses earlier published work related to the subject matter of the research and relevant points are presented.
- The main ideas developed in the paper should be laid out in detail using appropriate visual aids.
- The introduction usually finishes with a statement indicating the need to test the idea in an empirical evaluation, setting the stage for the method section.

8.2.5 Method

- Method section of a research paper explains how the experiment was designed and carried out
- Also called Methodology, Evaluation, Experiment, User Study, etc.
- It should be written straightforwardly, like a recipe, with simple and clear language
- Reproducibility is crucial, the reader should be able to reproduce the research after reading the method section
- Predictability is important to quickly find key points and results
- Conventionally, the method section is divided into the following subsections in the given order:
    1. Participants
    2. Apparatus
    3. Procedure
    4. Design

8.2.5.1 Participants

- The participants section of the method states the number of participants and how they were selected.
- Demographic information such as age, gender, and related experience is provided, and other details might be useful depending on the task and apparatus used in the experiment.
- The information on participants identifies the population of people used and to which the results apply.
- The population pool can be limited to people with certain skills or attributes if it makes sense for the research, but the results may not generalize to other people.
- The participants section is usually short, but if a property of the user is an independent variable, then it is often more detailed.
- An example is given where the participants section is quite detailed in an experiment where the independent variable was "expertise in judging websites for accessibility."

8.2.5.2 Apparatus

- The Apparatus section describes the hardware and software used in the experiment.
- It's important to give all the necessary details for replicating the apparatus.
- Mention only relevant details, not needless ones.
- If finger input on a touchscreen was used, the make and model of the touchscreen should be given along with other relevant details like screen resolution, screen size, sampling rate, etc.
- If a custom interface was used, the development language and other relevant details should be given.
- Providing screen snaps or photos of the interface is useful.
- Referring to earlier material is reasonable if comprehensive details of the interface were disclosed prior to the method section.

8.2.5.3 Procedure

- The procedure section describes what happened with each participant.
- It states the instructions given and whether demonstration or practice was used.
- If participants completed a questionnaire before or after testing, it is indicated in this section.
- It is common to indicate if participants were paid, volunteered without pay, or were required to participate.
- Participants are tested over multiple conditions and trials, which are described in detail.
- The section includes a specific task that participants performed, along with its definition and goal.
- Timing, errors, and error correction are defined precisely.
- Rest breaks, total testing time, and other relevant details are also included.
- A screen snap of the interface and a photograph of a participant performing the task can be useful.

8.2.5.4 Design

- The design section summarizes the experiment's independent variables (factors and levels), dependent variables (measures and units), or other relevant details.
- A statement such as "The experiment was a 3 × 2 within-subjects design" is often used to begin the design section.
- If counterbalancing was used, the way the conditions were administered to participants should be stated.
- The design section should be thorough and clear to ensure research reproducibility.
- The total number of trials administered in the experiment should be indicated, which can be calculated using the number of participants and variables and repetitions in the study.

8.2.6 Results and discussion

- Results and discussion are often combined in HCI experiments.
- Method, Results, and Discussion headings are at the same level.
- Mean task completion time for Method A was 2.7 seconds, and for Method B was 3.0 seconds (9.1% slower).
- Outliers and data collection problems should be stated and filtered data details provided.
- Results are commonly organized by dependent measures, starting with the most important variable, such as speed or task completion time, followed by accuracy and other dependent variables.
- Effect size is typically given as an absolute difference or as a relative difference.
- The results need to be explained through discussion, explaining the differences in the measurements across experimental conditions.
- Finding a real difference is often the goal of the research.
- ANOVA tests are used to test for real differences.
- The statistical test is just the supporting evidence.
- Experimental software logs a considerable amount of data, but only relevant results should be presented.
- Visuals like bar charts, line graphs, or 3D plots make results interesting.
- The results should be compared to prior research and feedback from participants included and discussed.

8.2.7 Conclusion

The conclusion summarizes what you did, restates the important findings, and restates the contribution. It is common to identify topics for future work, although developing new ideas is to be avoided in the concluding section.
The conclusion is often followed by an acknowledgment thanking funding sources as well as people who assisted in some way with the research.

8.2.8 References

- The last section of a research paper is the reference list.
- The reference list includes the full bibliographic details of cited papers.
- Only papers cited in the paper are included in the reference list.
- References should be formatted according to the conference or journal's requirements.
- Extraneous factors like static, noise, and distractions may affect the listening experience of music and the reading experience of a research paper.
- Presenting creative and well-executed research is not enough; the story must be properly assembled and presented.
- The author provides suggestions on preparing a manuscript in a straightforward style with simple and clear visuals.
- The research paper is a product of scholarship, not a marketing brochure.

8.3.1 Formatting

- Formatting is about small details, such as punctuation, capitalization, and italics, which are important for clarity and flow
- Reviewers may critique a paper based on confusing presentation or poor formatting, which could affect their opinion of the research
- The Publication Manual of the American Psychological Association (APA) is recommended for formatting rules
- The manual has rules for small details like punctuation and broader details like capitalization and spelling
- Dictionaries are a good source for spelling and capitalization rules
- Conferences provide template files for formatting papers, which embed many of the formatting requirements and provide guidance on aspects of the paper under the author's control

8.3.2 Citations and references

Citations and references are the connections that tie research together. Research submissions (e.g., for CHI) have formatting requirements for citations and refer- ence lists. This is one area where HCI conferences generally deviate from the APA guidelines. The following is a quick view into the formatting of citations and refer- ences for conference submissions to CHI and many other HCI conferences.

8.3.2.1 Reference list

- Figure 8.6 illustrates the formatting for five common types of publications in reference lists.
- References are numbered.
- References are ordered alphabetically by the first author's surname.
- The surname comes first, followed by a comma, then the initials for the given names. Include a space between the initials if there is more than one.
- For the title of the publication, only capitalize the first word, the first word in a secondary title (e.g., after a colon), and proper nouns.
- Always include the year. Substitute "in press" for accepted but not-yet-published papers.
- Always include page numbers (except for complete books or web pages).
- The name of the publication is set in italics with all keywords capitalized.
- For journal publications, include the volume number in italics.
- If space permits, use the full name for conferences and journals. If space is tight, use abbreviated names for conferences and journals. Do not mix full and abbreviated names; use one style or the other. If using abbreviated names, be consistent.
- Give the location and name of the publisher for conference papers and books.
- Use align left (ragged right) for the reference list.
- Only include works that are cited in papers.
- Study and imitate!
- Be consistent.

Other points:

- Citing web pages or documents downloaded from the Internet as the primary source for research is discouraged.
- If a relevant source is only available on the web or as a downloadable document, then it is reasonable to cite it.
- There are not yet standardized rules for citing and referencing these sources.
- The example in Figure 8.6 includes four items: the source, the title (set in italics), the URL, and the date the document was accessed.

8.3.2.2 Citations

(see page 326)

8.3.3 Visual aids

- Visual aids are powerful tools for conveying ideas and results in experimental research.
- Charts, tables, graphs, drawings, and photographs are now fully in the digital domain, with sophisticated tools for editing and stylizing the presentation.
- Bar charts are usually the most appropriate format for presenting the effect of an independent variable on a dependent variable when the independent variable is a nominal-scale attribute.
- If the variable shown along the x-axis is continuous, then a line chart is appropriate.
- Well before the Results and Discussion section, visuals are an important adjunct to text.
- Visuals can clarify concepts about the research or the technical details of an apparatus.
- A powerful way to augment the procedure section of a research paper is through a photograph of a participant performing the experiment task.
- Summaries of Likert-scale questionnaire items may be presented as a visual chart to help the reader understand the results both in absolute and relative terms.
- The natural model performed best in Zhu et al.'s experiment on teleoperation with the independent variable being camera control model (manual, natural, auto) and the dependent variable being the number of rocks sunk in a specified time interval.
- Error bars are an important reminder to the reader of variability in human responses.

8.3.4 Writing for clarity

- The goal in writing a research paper is effective communication with clarity.
- Clarity comes with economy, which means saying what needs to be said and little else.
- Writing with clarity is a craft and an art, and succeeding at both with balance is a challenge.
- In scholarly writing, clarity trumps all.
- Four sources are recommended for improving scholarly writing and methodology in experimental research.

- The technique for transforming a loosely written manuscript into a succinct and clear research paper is to get rid of clutter.
- Redundant words should be eliminated, and every word in a sentence should contribute to the point.
- The initial draft of a manuscript should be relaxed, and editing should be done before submission.